Multimodal Environmental Interfaces: Discrete and Continuous Changes of Form, Light, and Color using Natural Modes of Expression

by

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Abstract

In this thesis, I defined and implemented a framework for design and evaluation of Multimodal Environmental Interfaces. Multimodal Environmental Interfaces allow users to control form, light, and color using natural modes of expression. The framework is defined by categorizing possible changes as *discrete* or *continuous*. Discrete and continuous properties of form, light, and color can be controlled by speech, gestures and facial expressions. In order to evaluate advantages and disadvantages of each of the modalities, I designed and conducted a series of experiments. I disproved my hypothesis that whereas discrete changes are easier to control with language, continuous changes are easier to control with gestures and facial expressions through a series of interactive prototypes. I proved my hypothesis that the perception of whether a gesture or a speech command feels intuitive is consistent among the majority of users.

Thesis Supervisor: Takehiko Nagakura
Title: Associate Professor of Design and Computation

Acknowledgments

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For the realization of the complex installation project, I would like to specially thank Chris Dewart. Chris, I am indebted for your generous help with installing my exhibit. By no means conventional, the task required a good deal of brainstorming, expertise, and hours of hard work. I would like to thank Jim Harrington for his support and trust. Jim, thank you very much for helping me to negotiate the use of ACTs Cube space as well as letting me to hang my installation in the Long Lounge. I promise I will take it down on time. I would like to thank Cynthia Stewart. Cynthia, your kind and supportive attitude has been invaluable. I would like to thank my friends, Victor Leung, for helping me to design the electronics, and teaching me how to solder, Jeff Trevino, for composing the music, Ben Golder, for help with the calibration of the

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Thesis Reader: Terry Knight
Title: Professor of Design and Computation

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Chapter 1

Introduction

1.1 Motivations for Multimodal Environmental Interfaces (MEI)

While the idea of electronically-connected smart homes opens up a palette of novel ways to control our habitat, it only explores a limited range of possibilities that digital technologies offer for the design of our environments. Smart homes contain a common array of household objects with common functionalities. However, with embedded digital controls, these objects can perform their functions in the ways that are more intelligent. In this thesis, I look at how digital technologies can not only create more intelligent environments but fundamentally change the ways in which we occupy and interact with our environments. This change implies both creating novel kinds of objects and novel ways of interacting with them.

Advancements in material sciences, fabrication, and human-computer interaction present a vast field of novel design opportunities. On the one hand, shape-changing materials, variable-property materials, digital fabrication methods, along with the ability to integrate electronics, challenge traditional ways of form making. On the other hand, advancements in the field of human-computer interaction yield many novel opportunities for how we can interact with physical matter.

With these two main strands of innovation in mind – materials and human-computer interaction – the question arises : how can designers meaningfully explore the implications of these innovations for our physical environment?

The approach taken in this thesis is, first, to subjectively identify main features of the physical environments that these innovations can allow for. The thesis makes an assumption that these environments should have dynamic behaviours, i.e that they can change and adapt, and that these behaviours should be triggered through interaction with people. Secondly, the thesis devises a framework for relating different kinds of changes to certain types of interaction.

When communicating with each other, we use multiple modalities to convey meaning. Bodily expressions such as speech, gestures and facial expressions are interdependent and provide various levels of information. Although these modalities constitute the diversity of our experiences, spatial environments are hardly informed by the expressive human body. Would it be possible to interact with our spatial environments in ways similar to those with which we interact with each other?

Our environments are generally static. However, if we assume that architectural space can transform, change and adapt, then inhabiting a space will mean something different from what it means today. Inhabiting a space will be similar to having a conversation. In a conversation, speech, gestures and facial expressions are necessary for communicating infinite shades of meaning. In spatial environments, these same modalities will allow us to define a change. I will look at how spatial forms, light, and colour can change in response to bodily expressions.

1.2 Description of MEI

Multimodal Environmental Interfaces (MEI) are spatial interfaces that take speech, gestures, or facial expressions as input. To define a framework for design and evaluation of Multimodal Environmental Interfaces, I narrow down the possible changes down to those in spatial form, in light, and in color. I further categorize changes as either *discrete* or *continuous*. Discrete change is categorical. Continuous change is gradual.

This framework forms the basis of Multimodal Environmental Interfaces. The framework rests on the assumption that through a series of user tests, it is possible to arrive at series of guidelines that can help relate modalities to changes. The criteria that is used for the evaluation of how well a modality is suited for controlling a change are *learnability*, *efficiency*, *safety*, and *feel*. These criteria are borrowed from Neilsen's usability evaluation framework for Graphic User Interface (GUI) design. I believe that if the framework continues to develop, its impact could be similar to GUI design frameworks, i.e that it could provide sets of useful guidelines for MEI designers.

1.3 An Overview of the Precedents for MEI

1.3.1 Multimodal Interfaces in Computer Science

Multimodal Interfaces allow users to interact with digital devices using more than one modality; this may include speech and gesture, speech and lip movements, or gaze and body movements. Multimodal Interfaces first appeared in the early 1980s and are an intriguing alternative to Graphic User Interfaces. These new kinds of interfaces will allow for more expressive and natural means of human-computer interaction. Systems that allow for multimodal input can be significantly easier to use. They can be used in a wider range of conditions by a broader spectrum of people.[8, p.1] In Computer Science, the design of Multimodal Interfaces is challenging from both the perspectives of systems design and user experience. Systems design required

for multimodal human-computer interaction is fundamentally different from the traditional GUI architectures. Whereas GUI architectures have one single event stream and are sequential in nature, multimodal systems handle multiple input streams in parallel. The ways in which multiple streams of information are integrated differentiates the design of Multimodal Interfaces into *feature fusion* and *semantic fusion* approaches. Before going into detail about the different approaches to multimodal integration, however, I would like to make a survey of the methods for recognizing gestures, speech and facial expression. These modalities by no means limit the scope of multimodal interaction, which often includes gaze tracking, lip motion, and emotion detection. However, it is speech, gestures, and facial expressions that are the subject of this thesis. I will, therefore, discuss them in greater detail.

Speech and language

Speech and language processing have been fundamental to computer science from its very onset. The close relationship between speech and thought made computer scientists think about what it would take for a machine to be 'intelligent'. The well known Turing test (Alan Turing 1950) claimed that a machine can be considered intelligent if, when having a conversation with a human, the human can not tell that he or she is talking to a machine.

Putting the question of machine intelligence aside, this thesis aims to understand how speech – among with other modalities such as gestures and facial expressions – can become a means of interaction with the physical environment. In order to address this question, I will first give a brief overview of various speech and language processing paradigms. I will highlight contemporary approaches to speech and language processing. Similarly, I will analyze gestures and facial expressions through the prism of computer science. The analysis will allow me to draw a comparison among computational approaches to processing various natural modes of expression.

Speech and language processing is a complex task that comprises many levels of understanding. Imagine for a second being in a country where the inhabitants speak a language you have never heard. What knowledge would you need to acquire to be able to understand or speak it? First of all you would need to break sounds into words, which involves a knowledge of *phonetics* and *phonology*. You would also need to know the meanings of words, i.e. *semantics*. Being able to decipher words and their meaning would not be enough, however. Without the knowledge of *syntax* – i.e. how words relate to each other – it would be difficult to understand the overall meaning of phrases and sentences. *Morphology* – knowing the components of words and how they change – may be less critical but is also important in language understanding. If you want to engage in a dialogue, then apart from knowing how to take turns and pause – i.e. rules of *discourse* – you might need to know other culture-specific nuances about having a conversation. All of the above, however, would still not be enough for you to engage in a meaningful conversation. When we talk to each other, we also make assumptions about the intentions and goals of the speaker. These assumptions impact the way we interpret the meaning of words, an aspect of laguage referred to as *pragmatics*.

Despite the complexity of the task, speech and language processing have advanced dramatically since the middle of the twentieth century. State-of-the-art systems achieve up to 92.8 percent accuracy in language understanding (according to Professor Robert Berwick at MIT). When compared to other types of human-computer interaction such as gestures, emotions, and facial expressions, speech and language understanding are significantly more advanced. The success of IBM's Watson, Apple's Siri, Google Translate, and spell correction programs is evident. Although these systems may not be perfect, they clearly demonstrate the many advantages of speech-based interaction with digital devices. Web-based question answering, conversational systems, grammar checking, and machine translation are all active areas of research.

Historical overview

At least four disciplines have been involved in the study of language: linguistics (computational linguistics), computer science (natural language processing), speech recognition (electrical engineering), and computational psycholinguistics (psychology). [5, p.9]

The mid-twentieth century was a turning point in the study of language. With the inception of computer science came the idea of using finite state machines to model languages. In 1956, Chomsky described Context-Free Grammars, a formal system for modeling language structure.

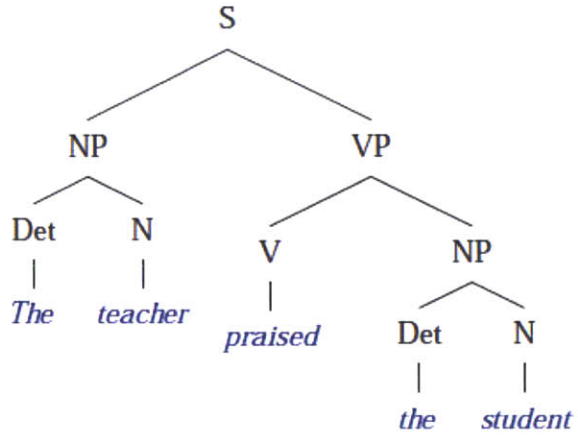


Figure 1-1: Context-free grammar parse tree. [2]

Context-free grammars belong to a larger field of *formal language theory* and are a way of describing finite-state languages with finite-state grammars. Another far-reaching work was done by Shannon, who was the first person to use probabilistic algorithms for speech and language processing. [5, p.10] In this same time period – 1946 – the sound spectrograph was invented. The invention was followed by seminal work in phonetics that allowed engineers to create the first speech recognizers. It was in 1952 at Bell Labs that the first statistical speech processing system was developed. It could recognize with 97-99 percent accuracy any of the 10 digits from one person. [5, p.10]

The early 1960s were significant because they marked a clear separation of two

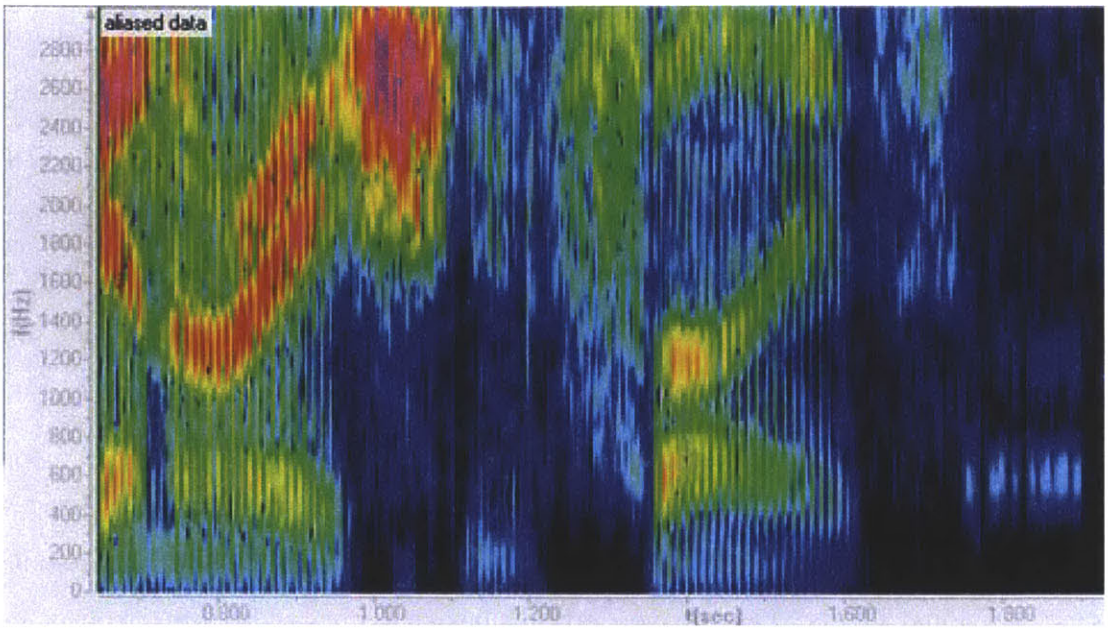


Figure 1-2: Spectrograph. [1]

paradigms: *stochastic* and *symbolic*. Whereas the stochastic paradigm was common among electrical engineers and statisticians, the symbolic paradigm dominated the emerging filed of Artificial Intelligence (AI). While the major focus in AI was on reasoning and logic (e.g. Logic Theorist, General Problem Solver), electrical engineering was focused on developing systems that could process text and speech (e.g. Bayesian text-recognition systems).

In the 1970s and 1980s, the field further subdivided into four major paradigms: *stochastic*, *logic-based*, *natural language understanding*, and *discourse modelling*. Each of these directions played a significant role in developing significantly more advanced and robust speech and language processing technologies. To name a few seminal works, IBM's Thomas J. Watson Research Centre and AT&T's Bell Laboratories pursued stochastic paradigms. Colmerauer and his colleagues worked on Q-systems and Metamorphosis Grammars, which largely contributed to logic-based models. [5, p. 11] In natural language understanding, the system SHRDLU developed by Winograd was a significant turning point. It clearly showed that syntactic parsing had been mastered well enough that the user could interact with a toy world of primitive

objects by asking such complex questions as 'Which cube is sitting on the table? Will you stack up both of the red blocks and either a green cube or a pyramid? What does the box contain? [SHRDLU demo, http://hci.stanford.edu/winograd/shrdlu/] Discourse modeling began to approach the tasks of automatic reference resolution. For example, consider the following sentences: I got a new jacket. It is light and warm. The meaning of 'It' follows from the first sentence. For a computer understanding, references across sentences is a non-trivial task.

In the mid and late 1990s, all the methods developed for parsing are enhanced with probabilities. For example, Probabilistic Context-free grammars significantly outperform traditional CFGs. Data-driven approaches that involve learning from a set of examples have become common place.

Since the beginning of the twenty-first century, there has been a growing fascination with machine-learning approaches. Machine learning is a branch of AI that studies algorithms that allow machines to learn from raw data. The interest in machine-learning for speech and language processing has been stimulated by two factors. Firstly, a wide range of high quality corpora has become available. Invaluable resources such as the Penn TreeBank (1993), PropBank (2005), and Penn Discourse Treebank (2004) are all well annotated with semantic and syntactic tags. These resources have allowed linguists to approach parsing using supervised learning techniques, which have proved successful. The second factor emerged from the downside of the first: creating good quality corpora is an incredibly expensive and tedious task. As a consequence, unsupervised learning approaches have emerged in an attempt to create machines that can learn from a very small set of observations.

Speech and language modeling, analysis, and recognition

Here I will use a quote from Daniel Jurafsky and James H. Martin that best describes the multifaceted nature of speech understanding:

'Speech and language technology relies on formal models, or representations of knowl-

edge of language at the levels of phonology and phonetics, morphology, syntax, semantics, pragmatics, and discourse. A number of formal models including state machines, formal rule systems, logic, and probabilistic models are used to capture this knowledge.' [5, p.10-11]

Gestures

Gestures have been studied for centuries. The first studies of gestures date back to the eighteenth century. Scientist looked at gestures in order to find clues to the origins of language and the nature of thought. By the end of the nineteenth century, however, the question about the origins of language was abandoned and the interest in gestures disappeared.[6, p.101] Whereas psychology was uninterested in gestures because they hardly shed any light on human subconsciousness, linguistics ignored them because its focus was on phonolgy and grammar. [6, p.101] In the mid twentieth century, however, the study of gestures was revived. Linguists became interested in building a theory of sign language, and psychologists began to pay more attention to higher-level mental processes. [6, p.101]

In the late twentieth century, the domain of computer science, specifically the field of human-computer interaction, defined a new dimension for discussing gestures. The notion of Gesture in computer science is different from its definition in psychology. Whereas psychological definitions view gestures as bodily expressions, in human - computer interaction a gesture is viewed as a sign. Human-computer interaction (HCI) domain understands a gesture as a sign or symbol. This view of a gesture makes it akin to a word in a language.

Taxonomy of gestures

Pavlovic [10, p.680] [11, p.7-8] proposed a useful taxonomy for HCI, which is driven by understanding gestures through their function. Firstly, the taxonomy separates gestures from unintentional movements. Secondly, it categorizes gestures into two types, *manipulative* and *communicative*. Manipulative gestures are hand and arm

movements that are used to manipulate objects in the physical world. These gestures occur as a result of our intent (i.e to move objects, to rotate, to shape, to deform) and our knowledge of the physical properties of the object which we want to manipulate. Unlike manipulative gestures, communicative gestures are more abstract. They operate as symbols and in real life are often accompanied by speech. [10, p. 680] Communicative gestures can be further divided into *acts* and *symbols*.

Acts are gestures that are tightly coupled with the intended interpretation. Acts can be classified into *mimetic* and *deictic*. An example of a mimetic gesture can be an instructor showing how to serve a tennis ball without any equipment. In this case, the instructor mimics a good serve but focuses purely on the body movement. Deictic gesture is simply pointing. According to Quek [11, p.7-8], when speaking about computer input there are three meaningful ways to distinguish deictic gestures: *specific*, *generic*, and *metonymic*. Specific deictic gestures occur when a subject points to an object in order to select it or point to a location. For example, clicking on an icon or moving a file to a new folder are deictic gestures. Generic deictic gestures are used to classify an object as belonging to a certain category. Metonymic deictic gestures occur when a user points to an object in order to define a class it belongs to. Selecting dumplings to signify Chinese cuisine is an example of metonymic deictic gesture.

Symbols are gestures that are abstract in their nature. With symbolic gestures it is often impossible to know what a gesture means without prior knowledge. Most gestures in Sign Languages are symbolic and it is difficult to guess their meaning without any additional information. According to Quek, symbolic gestures can be *referential* or *modalizing*. An example of a referential gesture would be touching one's wrist to show that there is very little time left. Modalizing gestures often co-occur with speech and provide additional layers of information. [11, p.9] For example, when one starts giving a presentation he or she might ask everyone to turn off their phones while at the same time putting a finger against his or her lips to indicate silence.

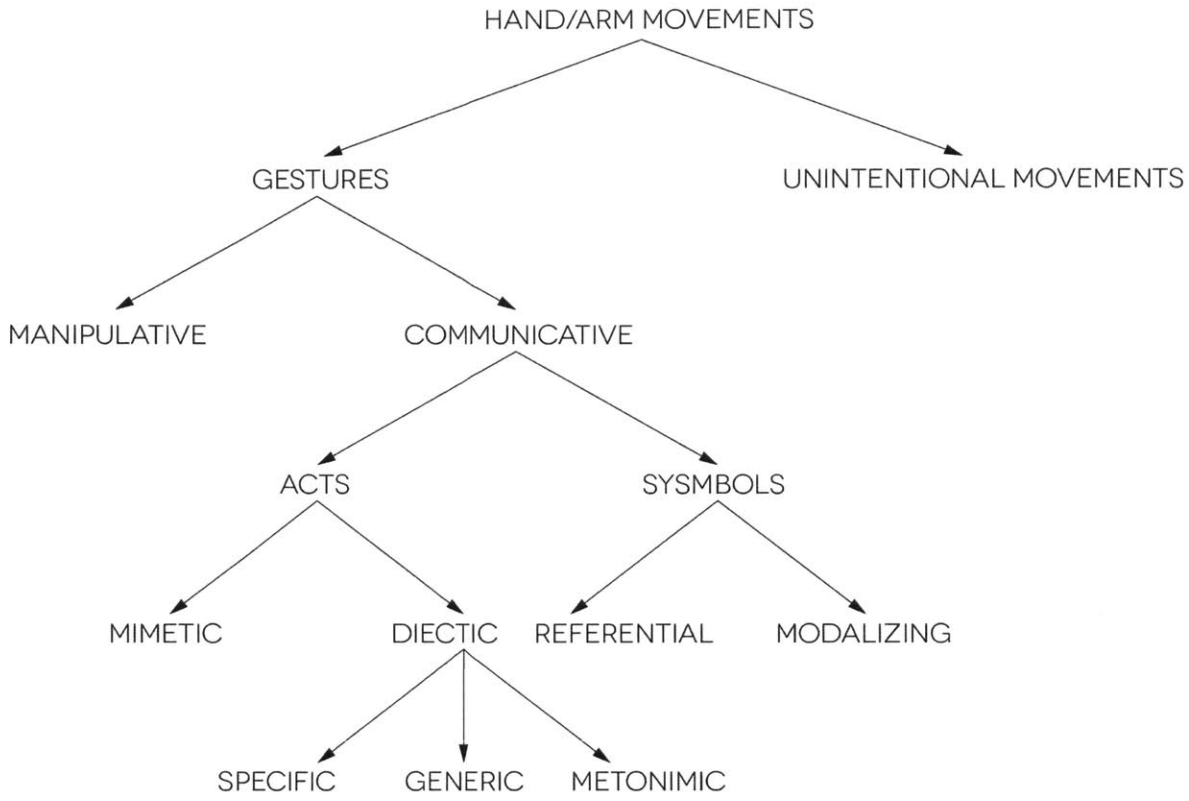


Figure 1-3: 'Taxonomy of Gestures' [10, p.680].

Gesture modeling, analysis, and recognition

While in real life the variety of gestures and their meanings is extraordinary, current computational systems focus on a limited range of gestures: pointing, wrist movements to signify rotation and location of objects in virtual environments, and single-handed spatial arm movements that create definite paths or shapes.

Computational systems that allow for gestural interaction integrate three modules: modeling of gestures, analysis of gestures, and gesture recognition.

Modeling of gestures

It is important to find the right way of representing gestures in order to efficiently translate raw video stream data into accepted representation format, and to com-

pare sample gestures with input. There are two approaches to gesture modeling: *appearance based* and *3D model based.* 3D model based approaches are generally more computationally intensive; however, they allow the recognition of gestures that are spatially complex. They can be further categorized into *volumetric models* and *skeletal models.* Volumetric models can be either nurb surfaces describing the human body in very great detail or constructed from primitive geometry(e.g cylinders and spheres). This method of comparison is called *analysis-by-synthesis*, which, in its essence, is a process of parametric morphing of the virtual model until it fits the input image. Using primitive geometry instead significantly reduces computation time; however, the number of volumetric parameters that need to be evaluated is still immense. In order to reduce the number of volumetric parameters, skeletal models have been studied. These models represent hands and arms schematically, and body joints have limited degrees of freedom.

Multimodal Fusion

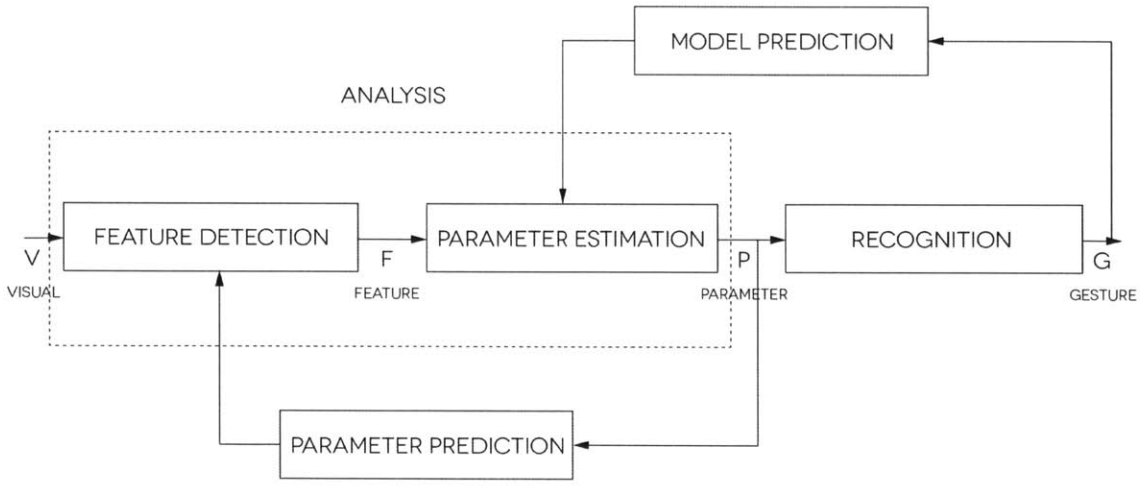


Figure 1-4: 'Analysis and recognition of gestures' [10, p.683].

Feature (Early) Fusion

Feature fusion is an approach in which modalities are integrated at an early stage of signal processing. This is particularly beneficial for systems in which input modali-

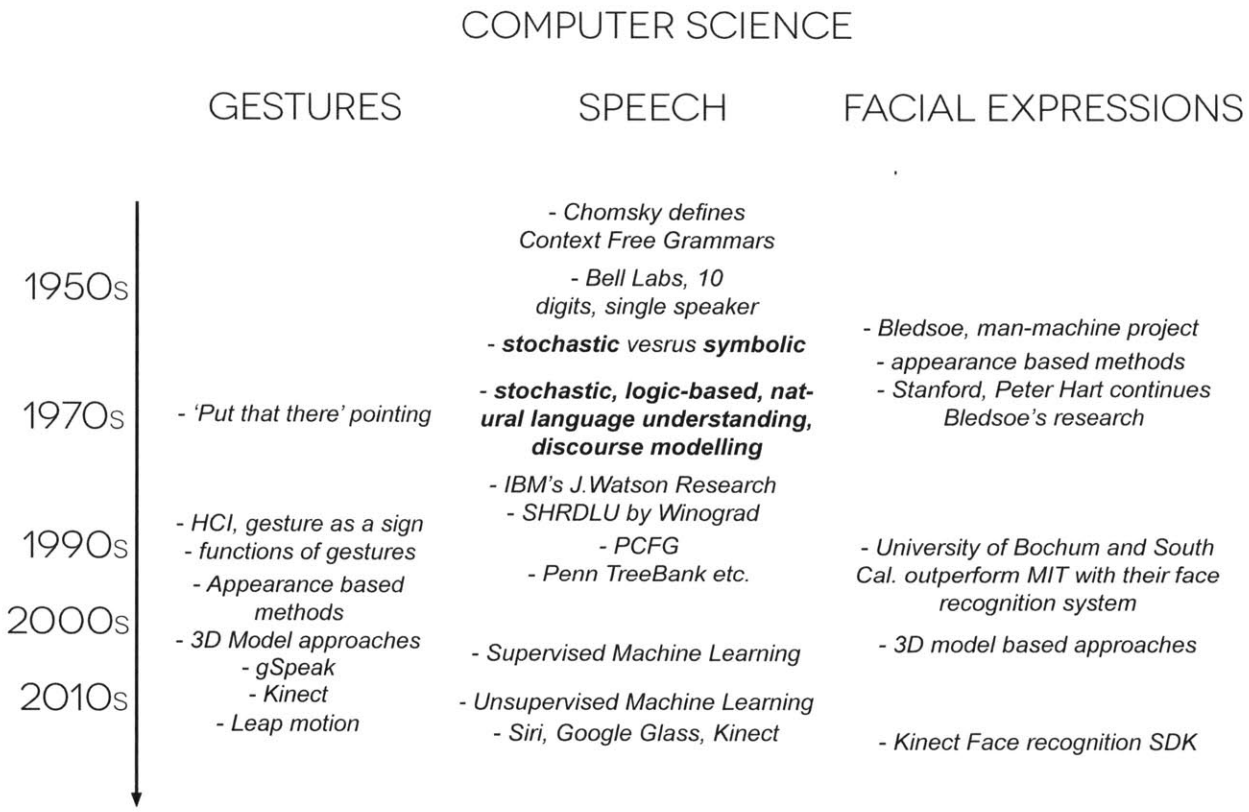


Figure 1-5: 'Framwork and Motivation'

ties almost coincide in time. A seminal example of a feature fusion system which I would like to discuss in more detail is a system developed by Pavlovic and his team in 1997. The system integrates two modalities, speech and gesture, at three distinct feature levels. Before describing the feature fusion approach, however, let me outline the architecture of the system. The system consists of three modules: a speech processing (visual), a gesture processing (auditory), and an integration module. [9, p. 121] The visual module receives input from a camera and processes the video stream. It comprises a feature estimator and a feature classifier. The feature estimator performs the following tasks: color-based segmentation, motion-based region tracking, and moment-based feature extraction. [9, p. 122]

Semantic (Late) Fusion

Semantic fusion is an approach in which modalities are integrated at a much later stage of signal processing. This approach is particularly beneficial when input modalities are asynchronous. Semantic fusion has a number of advantages. First of all, it allows modalities to be processed more or less autonomously. Secondly, recognizers are modality-specific and can therefore be trained using unimodal data. Thirdly, modalities can be easily added or removed without making substantial changes to the system's architecture.

1.3.2 The Architecture Machine Group

The Architecture Machine Group pioneered what we call today Multimodal Human-Computer Interaction. The seminal work 'Put that There' appeared in the 1970s. In this work it was first shown how a person could draw shapes on a screen using pointing gestures and speech. By using speech the user could define a type of a shape and its color; by pointing the user could indicate where the shape should be drawn. The project was further continued and evolved into an interactive placement of ships on the map of the Caribbean islands.

1.3.3 Interactive Design in Architecture and Art

For many years, there has been a fascination in the field of architecture with buildings that can physically transform and adapt. Realization of the idea took many different forms: from embedding motor controls into building components (e.g. transformable roof structures or motor actuated window shades) to augmentation of architecture with digital projection. [3, p. 3] Motivations for dynamic and responsive architecture are diverse and it is often difficult to draw a clear and continuous path of the evolution of ideas. Nevertheless, there have emerged distinct ways of thinking about transformation and adaptability in architecture. I systematically outline historic examples below.



Figure 1-6: 'Put-that-there'

History overview

I categorize precedents into two main groups:

1) Architectures than can physically transform and through that transformation reveal novel formal, pragmatic and cultural possibilities I refer to as Kinetic Architecture. Examples of Kinetic Architecture challenge traditional architectural elements and their function by incorporating digital motion controls and at times novel materials, and digital projection. Designers of Kinetic architecture often argue that their aim is not simply to solve problems but to create novel culturally meaningful experiences. Some of the examples of the work in this domain are Zaha Hadids 'Parametric Space' and Mark Goulthourpe's 'HypoSurface'.

2) Architectures than can physically transform and through that transformation improve ergonomics of our building I refer to as Smart Homes. Designers of Smart Homes accept traditional architectural elements as they are but enhance them with

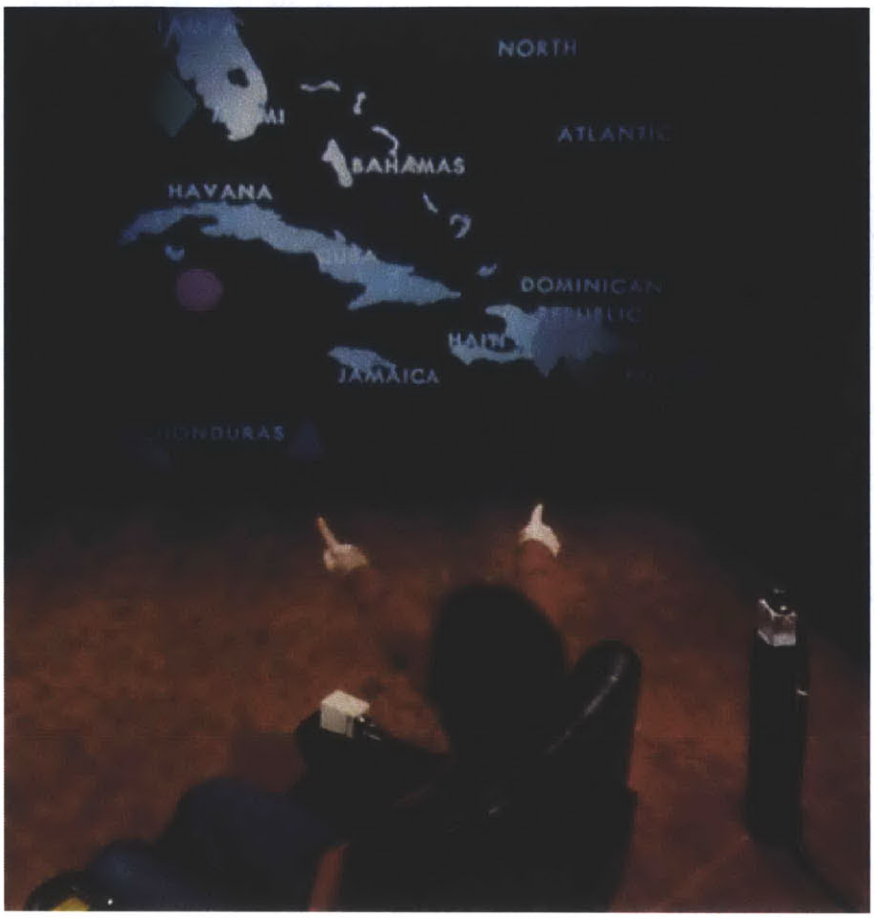


Figure 1-7: 'Put-that-there'

digital motor controls and electronic devices to control temperature, light, and humidity. For example, a regular tabletop that can be positioned at different heights or walls that can slide to create different spatial arrangements might be elements of a Smart Home. The argument for Smart Homes lies in their customization, efficiency, and their ability to provide healthier living conditions.

In both categories I do not limit spatial changes to motor actuated physical reconfigurations. Along with motor actuated changes, these architectures can also embed shape or state-changing materials, and digital projection.

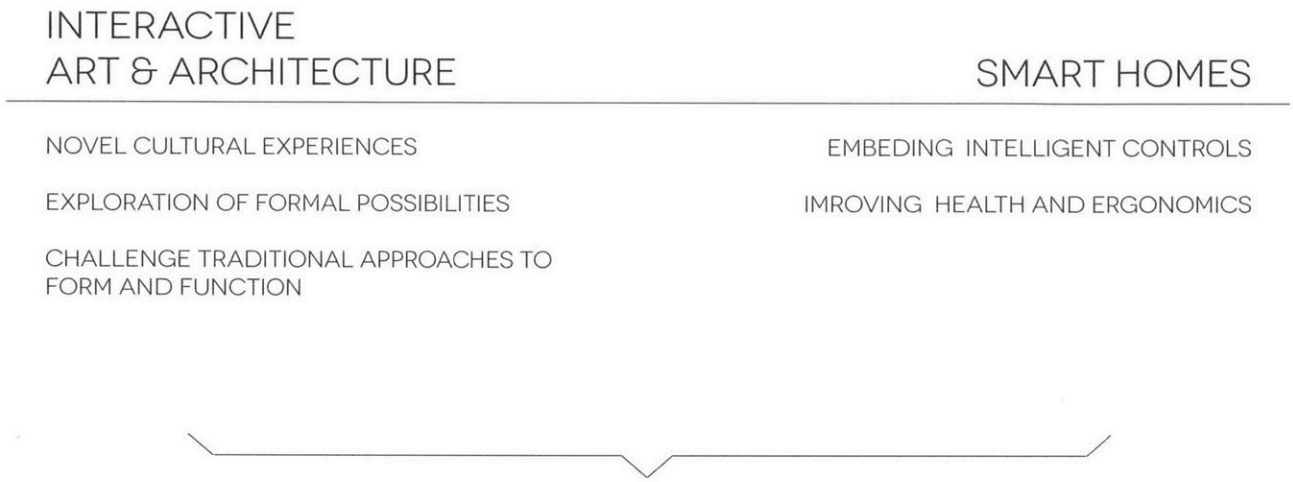


Figure 1-8: 'Framwork and Motivation'

Chapter 2

A Framework for Design and Evaluation of MEI

In Chapter I, I described the history and current trends in two distinct domains: Multimodal Interfaces in Computer Science and Interactive design in Architecture and Art. In this chapter, I will define Multimodal Environmental Interfaces (MEI) In the context of these two domains. I will explain what advantages they offer for our spatial environments. I will describe the challenges of designing MEI and, most importantly, propose a framework for design and evaluation of MEI.

Multimodal Environmental Interfaces (MEI) are spatial interfaces that allow users to interactively change spatial properties using natural modes of expression: speech, gestures, or facial expressions. The spatial properties that I am focusing on in this thesis are the physical transformation of space, light, and color. In future work on MEI, the set can be expanded to include thermal control, sound, and humidity.

2.1 Motivations for a Cross-disciplinary Approach

The reasons for a cross disciplinary approach are many. Firstly, this thesis aims to develop a framework for the design and evaluation of MEI; the closest analogy for this type of work is Jacob Neilsen's 10 Usability Heuristics for User Interface Design.

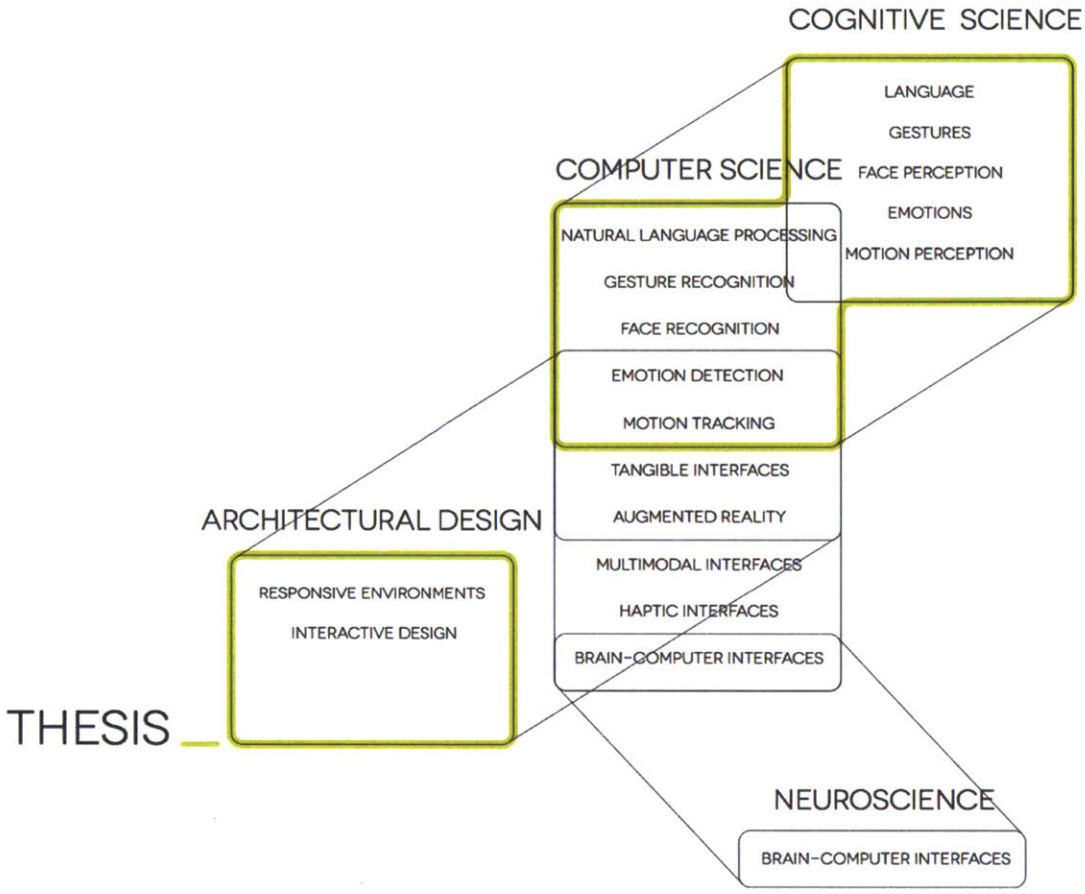


Figure 2-1: A Cross Disciplinary approach to MEI.

My thesis, therefore, borrows certain methods and lessons learned from developing design principles or guidelines for Graphic User Interfaces (GUI). Secondly, the methods required to recognize gestures, speech, and facial expressions are computationally very complex. It is commonly thought that designers do not necessarily invent new technologies but rather use them to fulfill their creative vision. Michael Fox, in his work *'Catching up with the past: a small contribution to a long history of interactive environments.'*, makes the following statement:

'Designing such [interactive] environments is not inventing after all, but appreciating and marshaling the technology that exists at any given time, and extrapolating it to suit an architectural vision. [4, p. 17]

A technological invention is therefore seen as a window into novel design opportunities. I claim that the current technologies for multimodal interaction evolved to serve applications that are significantly different from MEI and developing appropriate algorithms that seamlessly work with hardware is beyond the scope of a designer. There do not exist any 'plug and play' solutions to perform the functionalities that MEI would require. Developing MEI therefore is a fundamentally cross disciplinary work that at a minimum should involve Computer Scientists, Mechanical Engineers, and Electrical Engineers working alongside Designers from the very onset of a project.

It is somewhat intuitive to think that because we experience spatial environments through our body it would be most natural to control spatial changes using natural modes of expression. Speech, gesture, and facial expressions play an important role in how we understand the world and act upon it. Before going into greater detail about what constitutes the framework for design and evaluation of MEI, I will outline Neilsen's Usability Heuristics for GUI. The set of principles provides an important framework some parts of which will inform my arguments about the advantages and disadvantages of the natural modes of expression and multimodal interaction.

The field of User Interface Design has evolved a robust set of design principles and evaluation frameworks. One of the most common is heuristic evaluation frameworks is Jakob Nielsen's usability components. [7] Nielsen categorizes the goals of a good user interface into five categories: *learnability, efficiency, memorability, errors,* and *satisfaction.* He proposes that there are ten useful design heuristics, each of which helps to achieve one of the five goals.

10 Usability Heuristics:

Visibility of system status

The system should respond to the user in a timely manner by giving feedback that is easy to understand. We have all had the experience of becoming irritated by not knowing how long a web page would take to load. Giving the user information that

shows systems status without significant cognitive overload is the objective of this heuristic.

Match between system and the real world

This heuristic is used to improve learnability of an interface. If the system uses clues that are familiar to us from our every day experiences, then it is significantly easier to guess the underlying functionality. The principle often takes form of a visual metaphor. Apple successfully uses metaphors that are intuitive to help users grasp how their operating system works. Examples include such actions as dragging items to the trash, using trackpad gestures that are similar to manipulating physical objects, and so on. In its essence this heuristic implies that a system's functionality should be easily discoverable.

User control and freedom

The user should be able to navigate easily. This principle requires support of 'undo' and 'redo'. Accidental mistakes or slips are commonplace and should be easily erased to ensure efficient workflow.

Consistency and standards

If a task is common enough there should be a convention for handling it. The user should not wonder whether a different icon/word implies the same function or a different one. The feature of consistency contributes both to efficiency and learnability of an interface.

Error prevention

One of the easiest forms of error prevention is confirmation or safety dialogues. Although they can be extremely useful, they can pose a significant overhead for the user, especially if written in using technical jargon. Windows operating systems have been notorious for safety dialogues that are hard to understand for a non-technical user. The best practice is to minimize safety dialogues by decreasing a chance for an

error in the first place.

Recognition rather than recall

The user should have to remember as little as possible. Information on how to use the system should be easily available or/and be implicit in GUI design.

Flexibility and efficiency of use

Users often vary in their level of skill and experience. The Interface should be able to accommodate both professional and novice users by giving them flexibility. Not only do people have different skills, they also have different learning styles and ways of thinking. Giving users options on how to achieve a certain task allow them to discover their preferred ways of doing things.

Aesthetic and minimalist design

Redundancy should be avoided. Users should not be distracted by aesthetic features that are not meaningful or informative to them.

Help users recognize, diagnose, and recover from errors

If an error does happen the user should be able to diagnose the problem and find a solution as quickly and easily as possible.

Help and documentation

A good Interface Design tries to minimize the need to look up the documentation. This is not always possible, however, given the number of the features and complexity of a system. Help and documentation should be easy to access and navigate.

2.2 Framework Components

2.2.1 Natural Modes of Expression: Speech, Gestures, and Facial Expressions

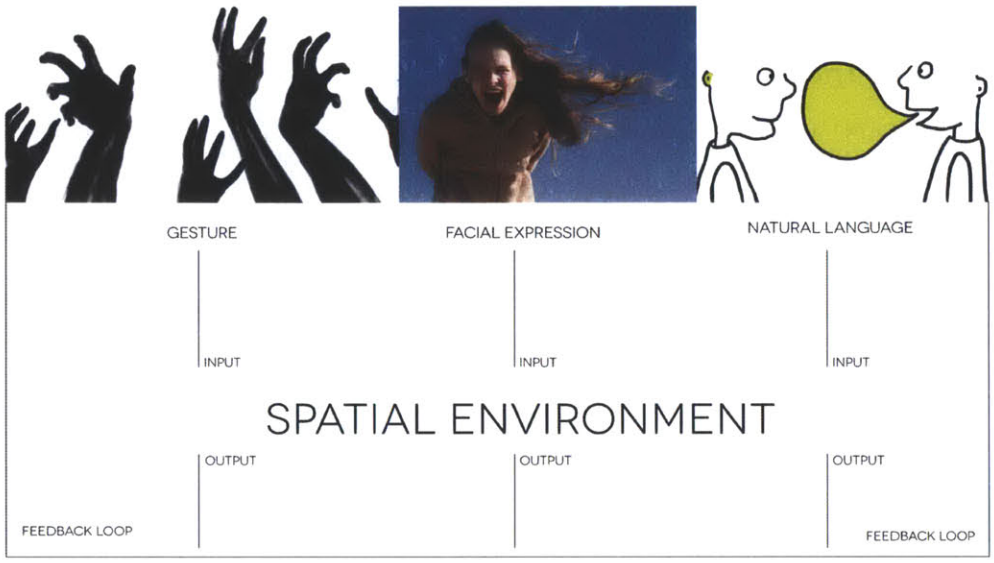


Figure 2-2: Natural Modes of Expression.

When we communicate with each other, speech, gestures, and facial expressions serve us as invaluable channels of information. While one modality may be sufficient for getting a message across, it is usually only when these modalities are perceived together that all shades of meaning are revealed in a conversation.

2.2.2 Discrete and Continuous Changes

The core idea of the framework is to categorize spatial changes into *discrete* and *continuous*. The reason why I think it is a meaningful distinction is because we tend to pay different levels of attention to the world that surrounds us. A lot of things remain unnoticeable until a certain threshold is reached and we become consciously aware about something. What is more, we seem to classify things and processes into distinct categories. This is particularly evident from the words in our languages. There are not that many words that describe the temperature of water. Water is

either categorized as 'cold,' 'warm,' 'body temperature,' or 'room temperature,' and perhaps in several other ways.

When it comes to spatial changes in order to interact with spatial environments in a simple and efficient manner, discrete changes or states are the immediate answer. For example, if I want to change an object from being a table to being a chandelier I would not want to go into all the details about the differences between the two. These objects represent two very different categories. When, on the other hand, I need a slightly different chandelier, then it might not be easy to communicate all the nuances by using discrete commands (like a language or a symbolic gesture). Instead a continuous, fluid gesture and immediate feedback seem most appropriate for a very nuanced differentiation.

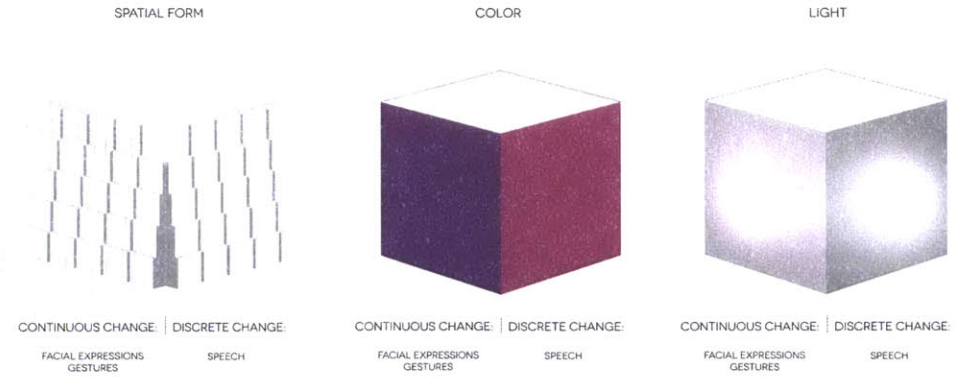


Figure 2-3: Discrete versus Continuous.

2.2.3 Changes in Form, Light, and Color

I focus on spatial changes such as light, color, and location. Each of these changes can be either discrete or continuous as shown in the diagrams above.

CHANGE	WAYS OF CONTROLLING THE CHANGE	EVALUATION
spatial form	speech, gestures, facial expressions	what is a preferred modality?
light	speech, gestures, facial expressions	how easy and natural does it feel ?
color	speech, gestures, facial expressions	what kind of sense of self does this type of interaction create?

Figure 2-4: Relationship matrix.

2.3 Transformative Space: Interactive Istallation Prototype

2.3.1 Transformative Space: Concept

Transformative Space Installation provides an example of what it could be like to interact with spatial environments using speech and gestures. The installation was sponsored by CAMIT Arts Grant.

Agenda

What is a meaningful relationship between archetypal spatial forms and digital information? Ceilings, chandeliers, stairs, walls, tables, chairs are looked at through the prism of the ephemeral new digital world. These entities are to form a spatial interface that negotiates the physicality of the real world and the infinite possibilities of the digital world. Language and gestures are to become the primary means of interaction with the spatial interface.

Installation

A 59x36 inch installation is composed of 48 translucent white cubes which are suspended from a ceiling. The cloud of cubes interactively reconfigures to form familiar objects, like a stair, a table, or a chandelier. Every cube is moved up and down by a small servo motor located in the ceiling. The suspension wire is transparent, making the cloud appear to float in the air, defying the forces of gravity. The lifting me-

chanical components are designed to be visible and create an industrial yet beautiful aesthetic. In contrast with the black mechanical parts, the cubes are weightless and ephemeral. The movement is activated through commands in natural language. For instance, when one would say 'a table' the structure would take a form that resembles a table.There are three possible configurations: a Table, a Chandelier and a Stair. Each of the configurations defines a state with its own functionality.

TABLE state:
When the installation is in a 'Table' state it functions as an image gallery. An image is projected from above onto the top face of each of the cubes. A projector is located in the ceiling and is connected to a laptop. A viewer can look through the image gallery by using a sliding hand gesture. Hand gesture recognition is performed using a Kinect sensor.

STAIR state:
The stair state reconfigures the cloud of cubes into a stepping pattern in such a way that formally establishes a dialogue with the surrounding space.

CHANDELIER state:
When in a 'Chandelier' state, the cubes move higher up and light up in different color patterns. Lighting is achieved by incorporating inside every cube a small micro controller, 4 LEDs and a distance sensor.

Architectural elements are often utilitarian and yet powerful means to convey architectural essence. The work is a subjective exploration of how these elements can be recast to find new meanings in the brave new digital world. Recent advances in computer science and sensor technologies, such as natural language processing and gesture recognition, are integrated into space and form making in order to create novel and socially meaningful spatial experiences.

Sharing the work

The installation is an invitation to a broad audience to question what new meanings familiar architectural objects can acquire in a world where digital information is no longer constrained to a surface or a mobile device. The installation was located in the ACT's Cube (Art Culture and Technolgy programme space) near the staircase, a location with perfect lighting conditions and spatial configuration. Currently the installation is located in the Long Lounge in the Department of Architecture. B-1 B-2 B-3

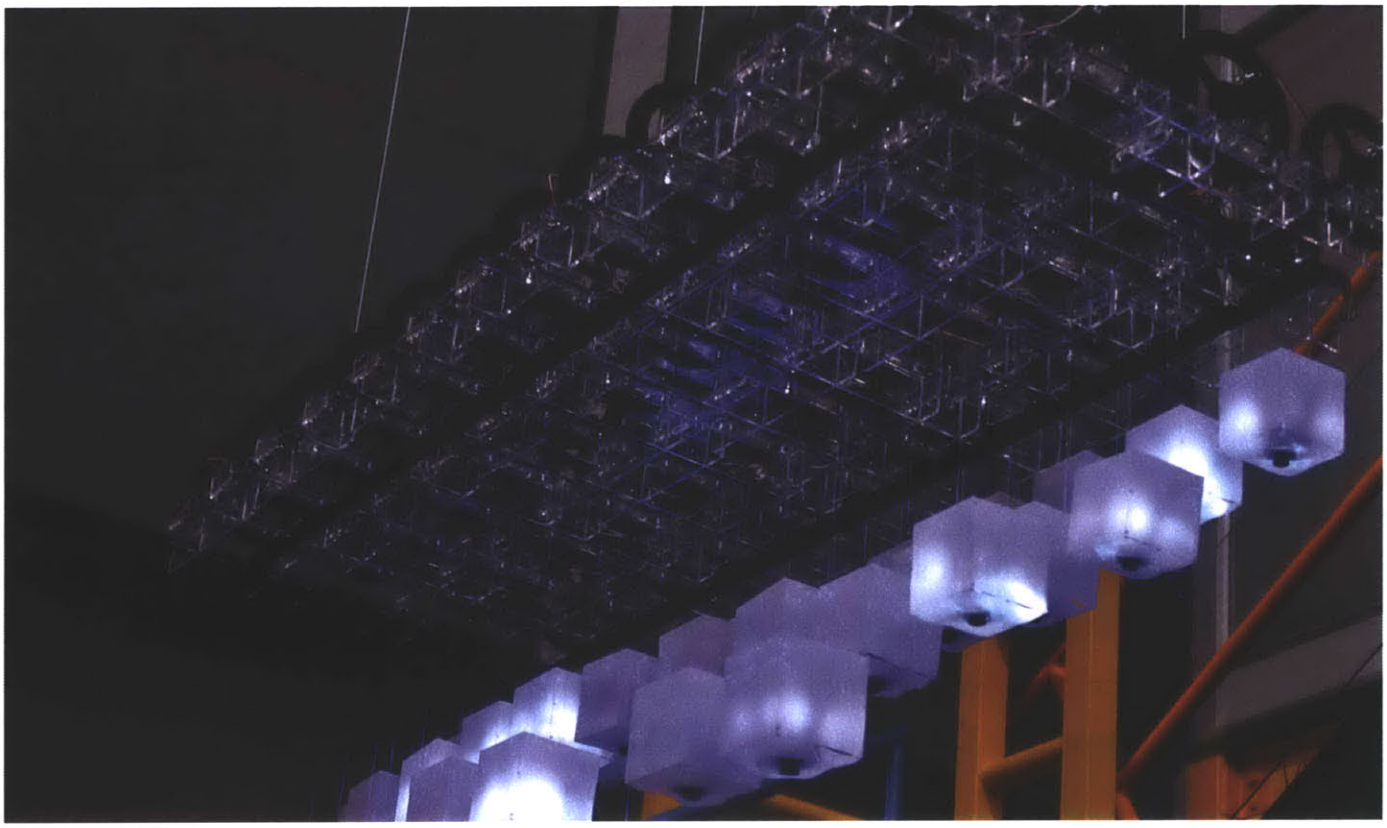


Figure 2-5: View of the Installation in Chandellier State in the ACT's Cube.

2.3.2 Transformative Space: Hardware Design

The installation consists of 48 cubes. Each cube is a standard unit that consists of a pulley with two fishing wires to move the cube up/down and to prevent it from

spinning. The wire is transparent, which makes it almost invisible from many view points. The pulleys are laser-cut out of black chipboard. Each pulley is attached to a *continuous rotation servo* motor. This type of a motor is a modified version of a traditional servo. Unlike a regular servo which has a limited 180 degrees rotation angle, continuous rotation servos can do multiple 360 degrees turns. However, position can not be controlled using angles directly, and instead position is a function of speed, time, and torque.

The matrix of cubes is divided into 2 by 4 racks, each of which is wired individually. There are therefore 8 racks which are plugged into a custom made power adaptor. Power adaptor that combines all of the 8 racks gets plugged into a power unit that supplies 5.5 V. The power unit is located on the very top of the installation and is powered from a regular outlet. I considered using an array of batteries which could be located on top and eliminate the need to run a cable from the top to connect to a regular power outlet; however, the number of batteries required and the length of time they could supply the motors with the right voltage was extremely short (about an hour). There was no other choice but to run a cable from the top part of the installation to the closest power outlet.

The motors are controlled with an Arduino Mega board, using up all of its digital pins. Although controlling Servo motors is generally a straightforward process when using Arduino Servo Libraries, in my case every motor responded to the same parameters slightly differently. I, therefore, had to do a complex calibration procedure for every cube. Nevertheless, even after calibration there could be an intolerance of up to one inch. The Arduino Mega board communicates with a computer wirelessly. To perform a wireless connection the board is equipped with a wireless shield and an Xbee transceiver Series 1. A Kinect is also plugged in to the same computer. With both the Kinect and the Arduino Mega connected to one computer, it is straightforward to relate mechanical motion and input from the Kinect.

Each cube has its own lights inside. Lights assembly consists of 4 LEDs, AdaFruit Gemma micro controller, Ultrasonic distance sensor, and a Lithium battery. Unfortunately there is no communication between the light and the computer. The lights therefore can not be controlled with either speech or gesture. In the installation the light intensity varies based on the height of the cube, which is determined by the distance sensor. The installation is programmed in such a way that in the 'Table' state the light are off. When the installation transitions from the 'Table' state into the 'Chandelier' state then lights gradually light up as the cubes are moving higher up. When the installation reaches the 'Chandelier' state then the light intensity becomes stable and is determined by the high of each of the cubes. B-4



Figure 2-6: Assembled light components that are located inside the cubes.

2.3.3 Transformative Space: Software Design

The core piece of software is written in C++ in Visual Studio 2010. The Arduino Mega board and AdaFruit Gemma are programmed using Arduino programming interface. Communication between the main application and the boards occurs over a serial port.

The primary function of the main application is to process multiple streams from the Kinect sensor audio data and skeletal tracking and to determine whether a physical motion event should be fired. If an audio event is signaled then the speech recognition engine evaluated if an utterance corresponds to any of the words listed in the grammar. I the case of the installation the grammar includes a 'Table', a 'Chandelier', and a 'Staircase'. This grammar can be easily expanded to include many words. The greater the number of words, however, the higher the chance that the system will make a mistake and start moving when movement is unwanted. Skeletal tracking stream is a basis for gesture recognition. Gesture recognition is performed using an algorithmic method. Algorithmic method implies that a gesture is defined through a series of parameters. How well these parameters get matched determines whether an even gets fired. There are two gestures that are implemented to operate the installation: a 'slide up and a 'slide down.

Arduino programs comprise patterns for mechanical motion and an interface for moving each of the cubes individually or at once.

2.3.4 Transformative Space: Challenges and Future Work

Having more reliable and precise motors would make the most difference for this project. The low torque of the motors used in the installation limited the number of materials that cubes could be made of the cubes had to be as light as possible. Although vellum paper proved to be a good solution, white plexiglass is a more attractive option from aesthetics perspective.

Another improvement could be made in the number of gestures the system can recognize.

It would also be interesting to integrate facial expression recognition into the process, especially that it can also be handled by the Kinect Sensor.

2.4 Experiment Design: User Experience

The design of the installation raised a number of questions about how multiple modalities should be used and integrated together. What defines a good gestures or verbal command? Which changes are better suited for gestural interaction and which are better suited for speech? What criteria does 'better' involve? Would the preferences be consistent across different users? To what degree would user preferences vary? Which features would be most valued by the users? How easy would it be to discover how the system works by simply interacting with it? In order to address these questions systematically, I designed a user experiment that looks at how users interact with a single component from the installation: a cube that can move up or down in response to speech or gesture and lights inside the cube that can also be controlled using the two modalities.

2.4.1 Hypothesis

The experiment is designed to test a set of qualitative hypotheses. The first hypothesis claims that users would prefer to use speech to define discrete states of the system. Continuous changes would be easier to control with gestures. The second hypothesis states that a perception of whether a gesture or a spoken command is intuitive and natural to use is consistent across the users.

2.4.2 Prototype and Questionaires

The experiment tests three assumptions:
1) whether discrete changes are easier to control with speech and continuous changes are easier to control with gestures.
2) the perception of whether a gesture or a spoken command is intuitive and natural to use is consistent across the users.
3) both encoded speech commands and encoded gestures are easily discoverable. In other words, if the users know what the system can do, they can learn how to control the system intuitively without the need for a manual.

The idea for the experiment is to first introduce the prototype to the users by explaining what it can do: *The prototype is a cube that can be moved up and down with either a speech command or a gesture. The cube can also light up and the intensity of light can be varied similarly by using speech or gesture.* After test subjects have seen the prototype and know what it can do, they are asked to accomplish a specific task by using a single modality. However, the subjects do not know which specific words and gestures are encoded for moving the cube up and down. The idea therefore is to record all the words and gestures the users try, compare them to the encoded commands and compare how consistent the guesses are among different users. Below I provide an outline for the experiment. The order is shifted for every new user test to eliminate bias.

PART 1: Spatial Location

Speech

Subject is asked to move a cube lower or higher using SPEECH.

Questions:

- Are the encoded words easily discoverable ?
- How consistent is the word choice among the users ?

Analysis: map the words used in a word cloud, scaling the words based on their usage.

Gestures

Questions:

- Are the encoded gestures easily discoverable ?
- How consistent is the choice of gestures among the users ?

Analysis: a video recording of the gestures made by the users with indication of how popular that gesture was.

PART 2: Light

Speech A

Subject is asked to turn light ON and OFF using SPEECH.

Questions:

- Are the encoded words easily discoverable ?
- How consistent is the word choice among the users ?

Analysis: map the words used in a word cloud, scaling the words based on their usage.

Speech B

Subject is asked to change the intensity of light using SPEECH.

Question:

- Are the encoded words easily discoverable ?
- How consistent is the word choice among the users ?

Analysis: map the words used in a word cloud, scaling the words based on their usage.

Gestures A

Subject is asked to turn light ON and OFF using GESTURES.

Question:

- Are the encoded words easily discoverable ?
- How consistent is the choice of gestures among the users ?

Analysis: a video recording of the gestures made by the users with indication of how popular that gesture was.

Gestures B

Subject is asked to change the intensity of light using GESTURES.

Question:

- Are the encoded words easily discoverable ?
- How consistent is the choice of gestures among the users ?

Analysis: a video recording of the gestures made by the users with indication of how popular that gesture was.

2.5 Experiment Implementation

The experiment is implemented using one element: a cube that is moved up and down by a pulley above it. The cube is fitted with light. Both the position of the cube and lighting can be controlled by either using speech or gesture. There are two Arduino Uno boards that control the system: one that is attached directly to a computer and one that is fitted inside the cube. The former board controls the servo motor. It can receive input from the Kinect sensor through a serial port. The latter board is located inside the cube and is used to control 4 LEDs. Similarly, it can receive information from the Kinect sensor; however, this time communication happens wirelessly. Wireless communication is achieved by using Xbee transceiver Series 1 and an Arduino wireless shield. Being able to send signal wirelessly in this case is crucial because it eliminates wires connecting the moving cube to the computer.

A Kinect sensor is used for speech and gesture recognition. The main program is written in C++ Visual Studio 2010. It receives two input streams from the Kinect: an audio stream and a skeletal stream. The system waits for either speech events or skeletal events. If a speech event is fired, then speech recognition is performed. If the word matches the predefined grammar then a signal is sent to Arduino and either mechanical motion or light become activated. Skeletal tracking events are activated at a much higher rate than speech. When Skeletal tracking is activated then Gesture recognition is performed. The gesture recognition module takes 4 joint 3D coordinates of the right hand and arm as input. The 3D coordinates have a time stamp on them. Gesture recognition is performed using an algorithmic approach. Given a sequence of the 3d coordinates and time stamps I evaluate deviation of motion in X, Y, Z planes. If the deviation is within the threshold defined in the predefined description of the gesture then I check the time stamps. If the time stamps also fall within the predefined thresholds then the gesture event is fired and a signal is sent to Arduino.

2.6 Experiment Data Collection

The data is collected from 10 users. Age of users range within the bounds of 22 years old and 36 years old. Among the 10 test subjects 7 were males and 3 were females. All the test subjects are currently students at MIT. At the beginning of each of the experiments the subjects were asked if they agree to participate in the experiment. A formal approval was taken from the subjects that the results of the experiment can be described in this thesis as well as future academic papers. The subjects were informed that they could interrupt the experiment, ask questions, give comments, and if for any reason they want to stop participating in the experiment then they were free to do so. At the end all of the subjects successfully went through all stages of the test.

2.7 Experiment Data Analysis

Data analysis includes a series of graphs that highlight various aspects of the experiment that I will discuss in greater detail below. The analysis also includes a series of quotes from the users.

In Figure 2-7 I analyze which words were used by the users in order to turn lights on and off as well as to change the intensity of light to brighter or dimmer. The X axis represents the words used by the users and the Y axis represents how many people used that word in an attempt to change the light. Encoded words for turning lights on/off were 'lights on' and 'lights off'. Encoded words for making lights brighter or lighter are 'brighter' and 'dimmer'. All ten users easily discovered the encoded words with the exception of 'dimmer'. Although the encoded words seemed to be intuitive to discover, a significant number of other expressions were also used, such as, 'more intense', 'less intense', 'less bright', 'stronger', 'up', and 'down'.

In Figure 2-8 I analyze which words were used to move the cube higher or lower. The X axis represents the words by the users and the Y axis represents the number of users that used that word. The encoded words, 'higher' and 'lower' did not appear to be the most common choices. All users preferred 'up' and 'down'. There was also a range of other expressions, including 'cube, move down', 'upwards', 'downwards', 'go up', and 'much lower', 'much higher'. In Figure 2-9 I analyze whether the gestures encoded for making light brighter or dimmer were easy to discover. The X axis represents the gestures that users tried. 'Slide up/down' means a hand is moved up/down vertically; 'flashing' refers to an arm opening and closing repeatedly; 'small rotation' is a circular arm motion; 'slide left/right' is a straight hand motion that is parallel to the floor. The Y axis represents the number of users who tried the gesture. The encoded gestures were 'slide up' and 'slide down'. The experiment showed that along with coming up with somewhat unexpected gestures, like 'flashing', the users showed equal preference to vertical and horizontal sliding motions.

Figures from 2-12 to 2-16 demonstrate evaluation of each of the interaction method in relationship to changes along 5 dimensions borrowed from Jakob Neilsens usability components for graphic user interface design and implementation: learnability, efficiency, memorability, errors, and feel. *Learnability* is measured by the umber of trials required to discover the encoded word or gesture. 1-2 trials is 100; 2-4 trials is 70-100; 4-6 trials is 50-70; 6-8 trials is 30-50; 8-10 trials is below 30. *Efficiency* is closely related to learnability. It is a measure of how quickly the users can achieve a desired outcome. Speech proved to be more efficient. *Memorability* means how well the users remember the discovered words ands gestures. There was no difference in this dimension among the modalities. Memorability was measured by asking the users to write down the commands and gestures they discovered after the experiment. Errors refers to how often a command will need to be either restated or a command would lead to an undesired outcome. Overall there were fewer errors for speech commands; however, the number of errors was still significant. *Feel* factor is evaluated by asking users to rate a modality for controlling a certain type of change after the experiment was complete. Interestingly, despite the fact that speech proved to be preferred over gestures in the majority of dimensions, gesture significantly outperformed speech on the dimension of 'feel'.

Subjective evaluation feedback:

The following comments came from subjective evaluation feedback: 'Certainly I would just use whichever worked more effectively and produced more immediate feedback from the device. Having a sense of control, irrelevant of the medium, is most important.'

'Saying things to a computer felt a little bit uncomfortable. I think in theory I would prefer gestures. But I wouldnt know when to tell it to start watching my gestures. How would I indicate that it should start watching my hand? It makes me think that it might make sense to have somewhat peculiar hand gestures.'

'The voice commands were more responsive, and the fact that they were discrete made it easier to know if they were actually working.'

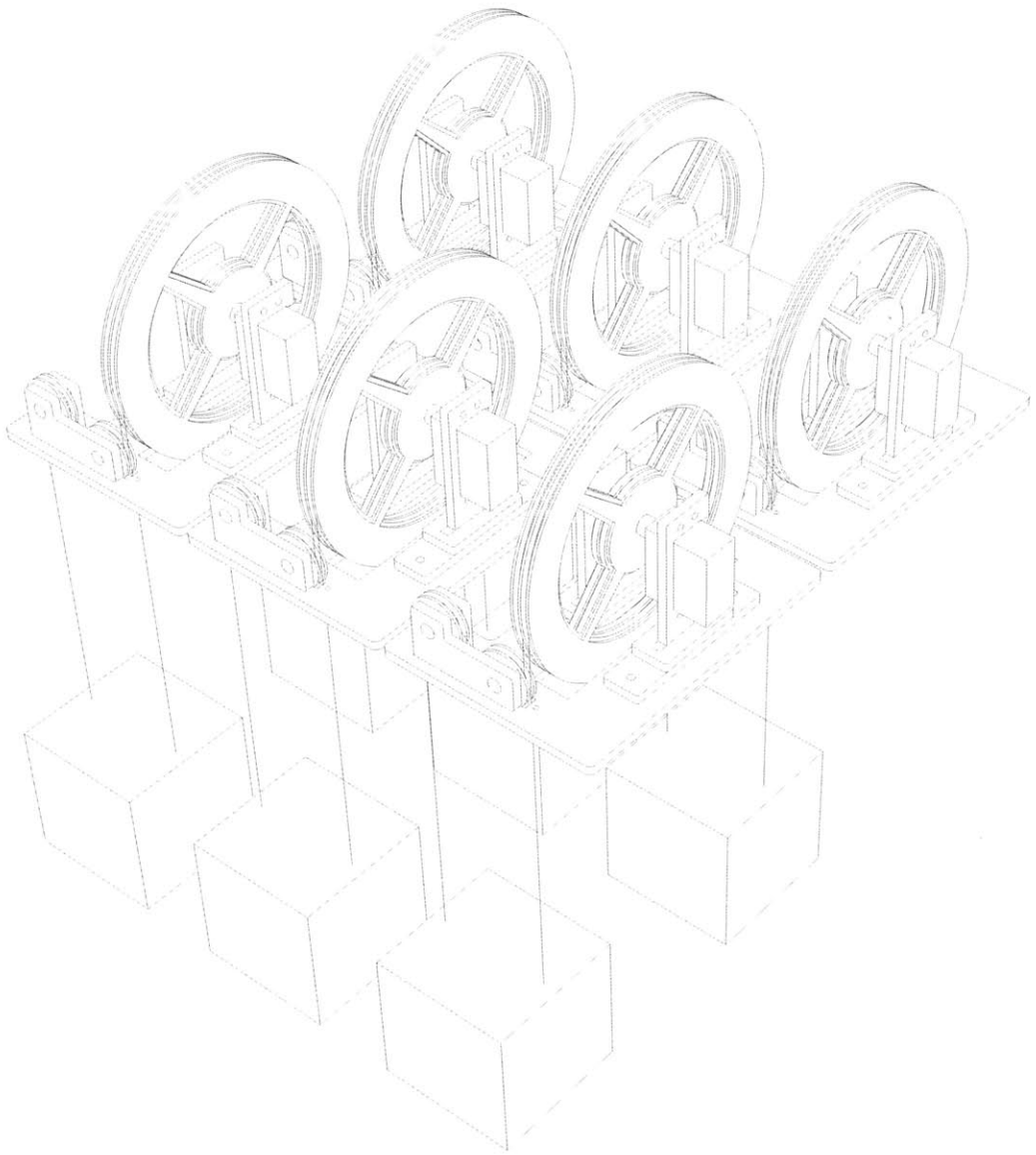


Figure 2-7: Perspective view of 6 components showing the pulley mechanism.

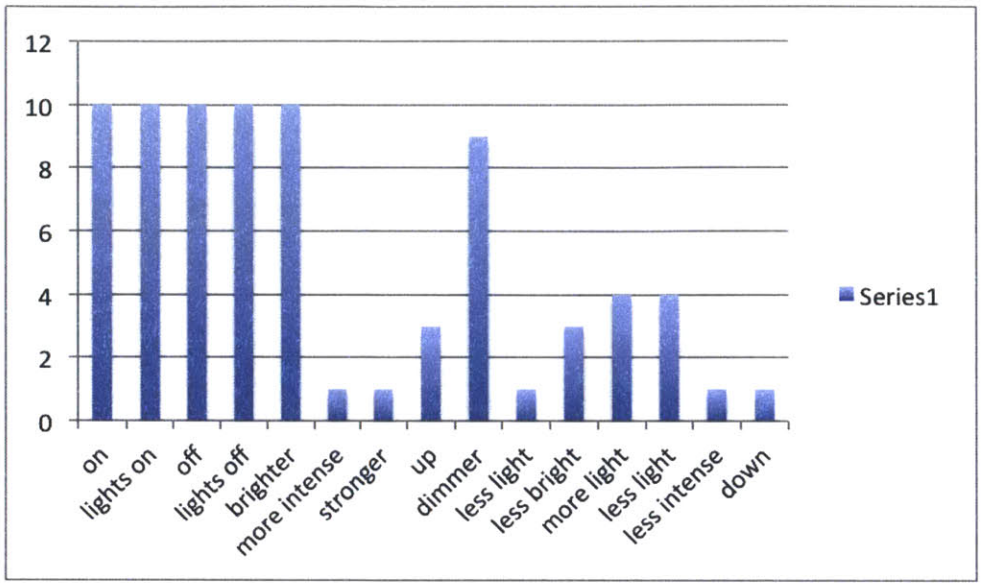


Figure 2-8: SpeechLightDiscoverability.

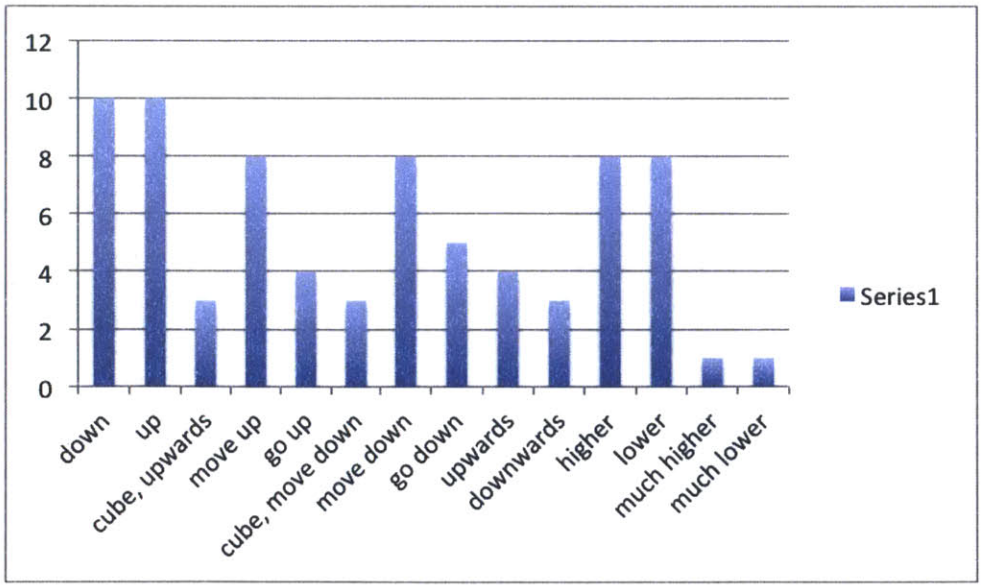


Figure 2-9: SpeechLocationDiscoverability.

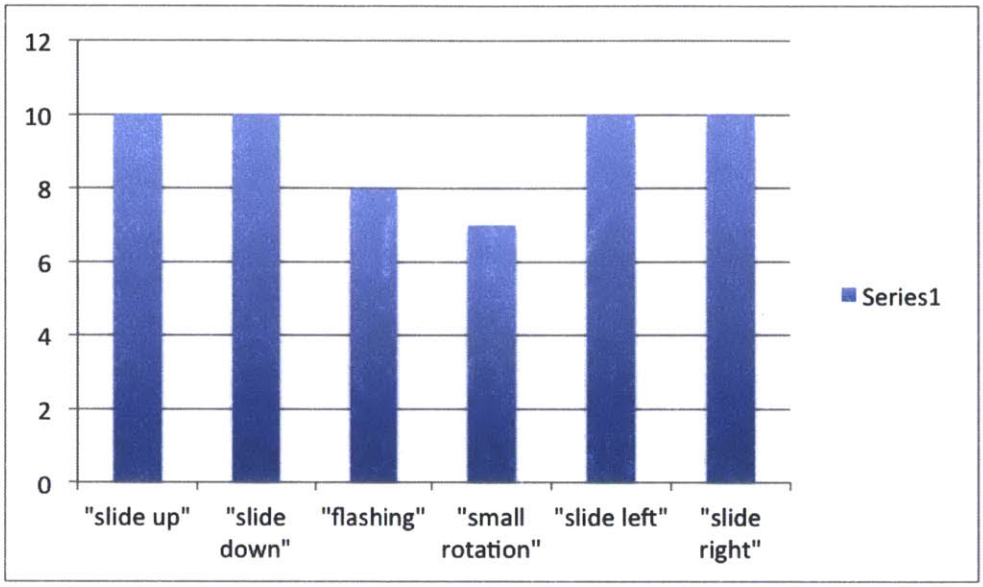


Figure 2-10: GestureLightDiscoverability.

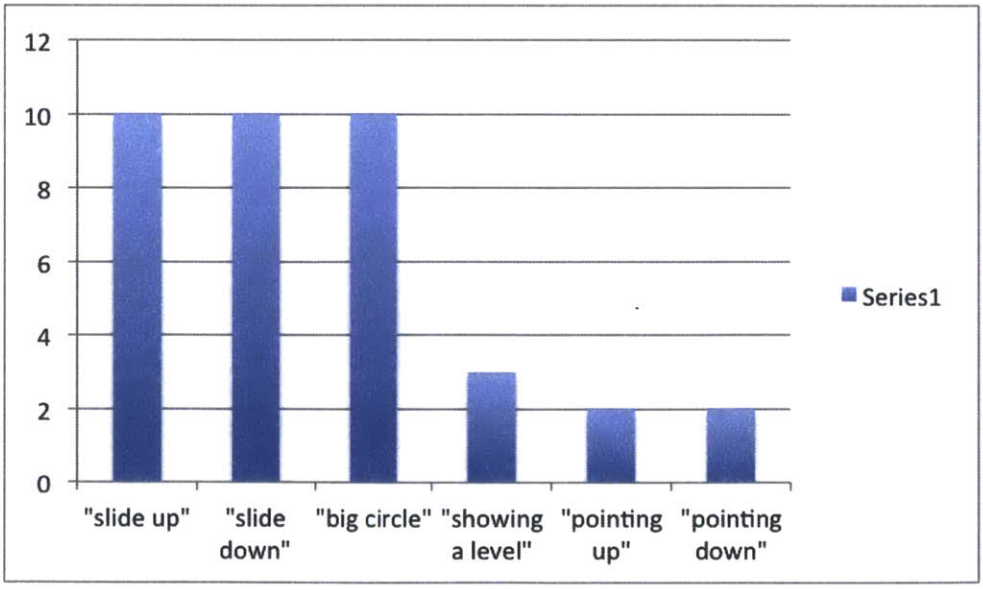


Figure 2-11: GestureLocationDiscoverability.

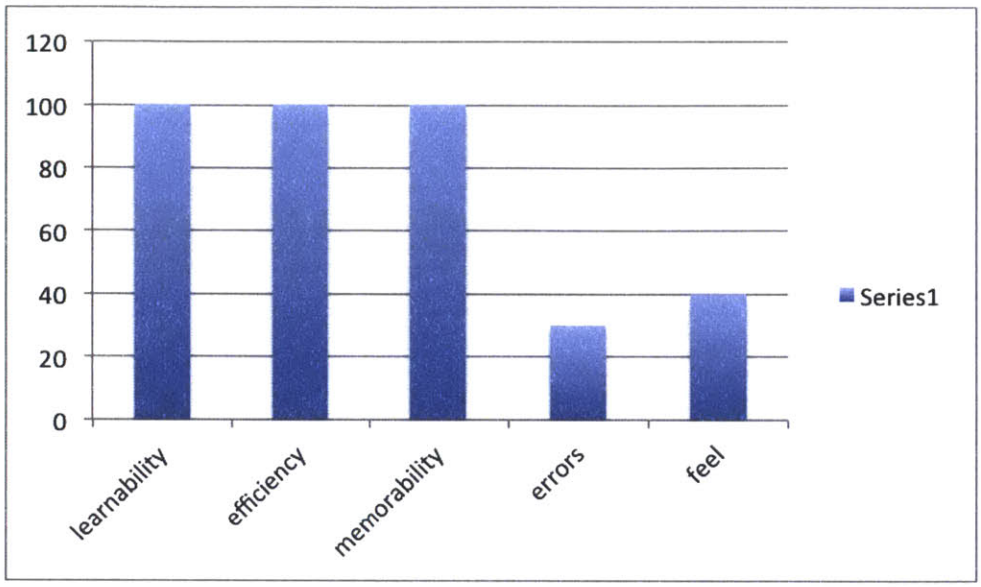


Figure 2-12: LightSpeechDiscrete.

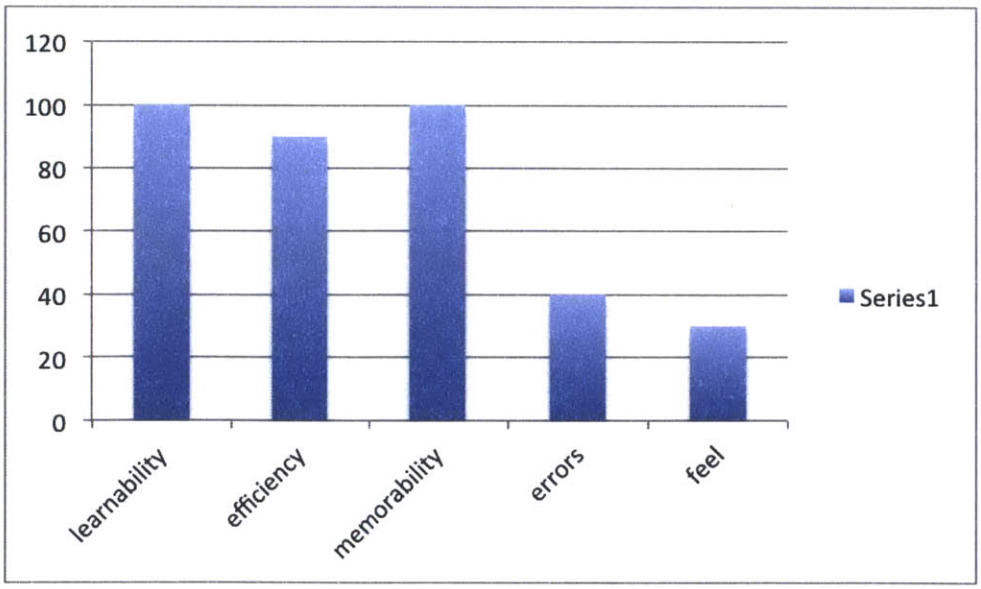


Figure 2-13: LightSpeechContinuous.

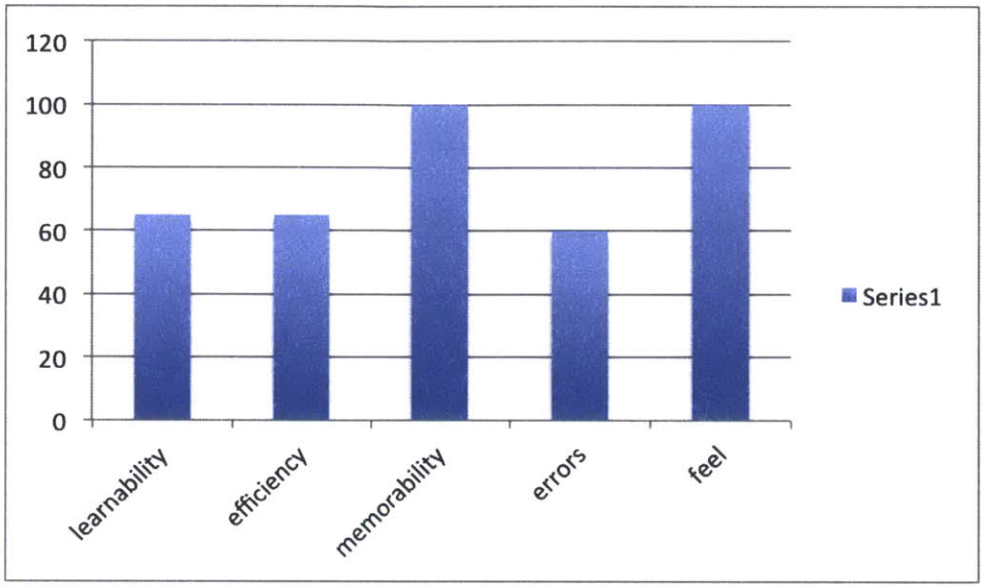


Figure 2-14: LightGestureContinuous.

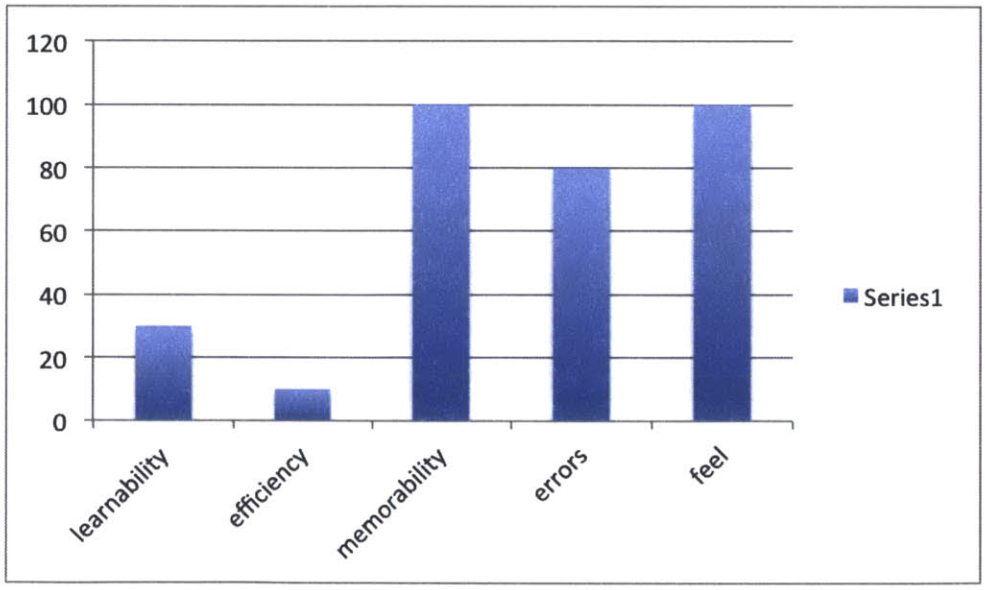


Figure 2-15: LightGestureDiscrete.

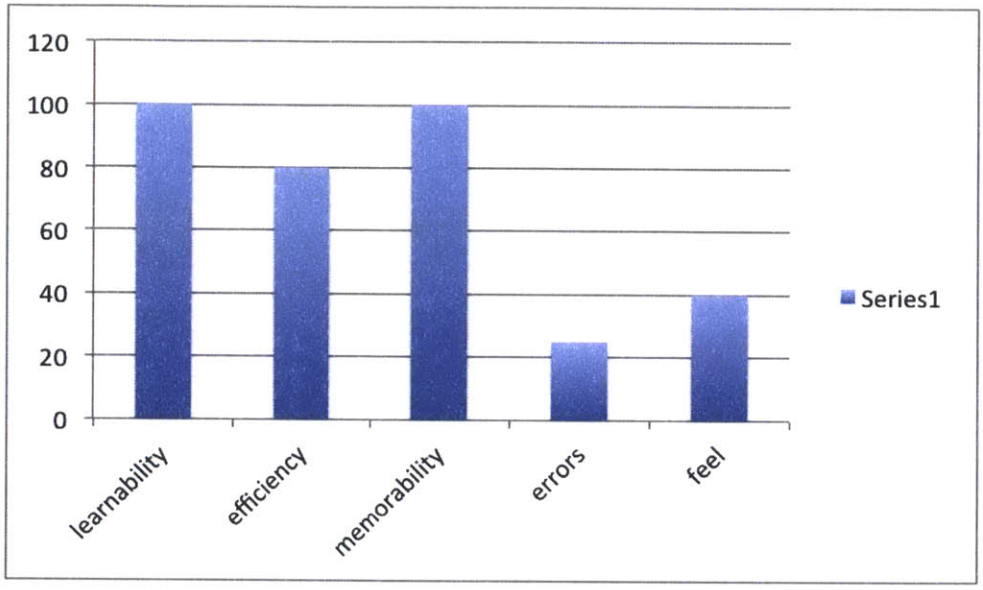


Figure 2-16: MotionSpeechContinuous.

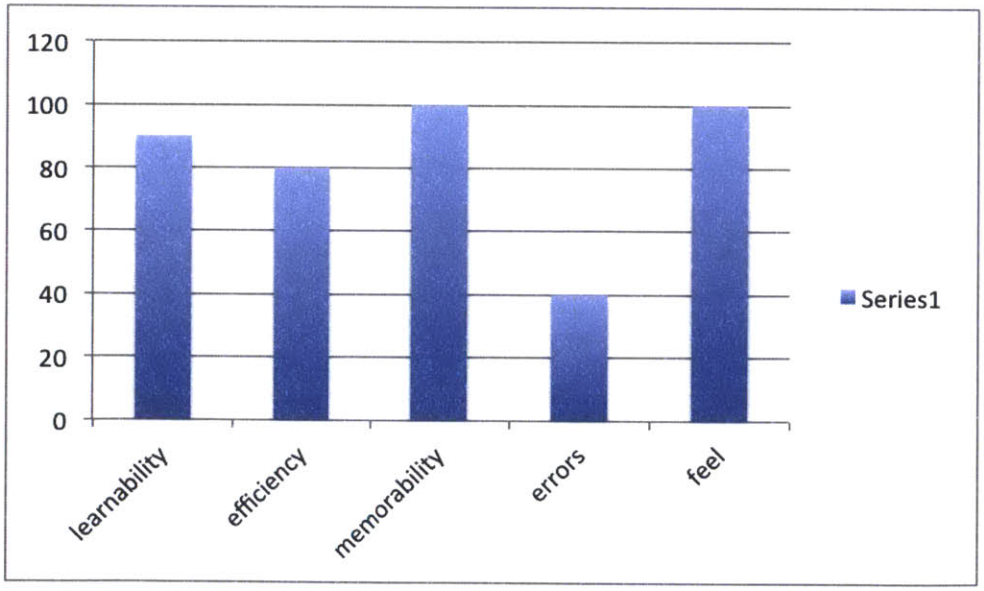


Figure 2-17: MotionGestureContinuous.

Chapter 3

Contributions

3.1 An Overview of the Contributions

- I created and defined the concept of Multimodal Environmental Interfaces (MEI).
- I proposed a framework for the design and evaluation of MEI.
- I developed a method for experimentation that allowed me to individually test each of the changes in relationship to each of the modalities.
- I demonstrated how such spatial properties as form, light, and color can be interactively changed using speech, gestures, and facial expressions.
- I built a series of prototypes through which I evaluated the advantages and disadvantages of each of the modalities for interacting with spatial environments.
- In order to build the prototypes, I developed custom software and hardware. I used currently available sensor technologies – i.e. the Kinect sensor – to perform gesture, speech, and face recognition.
- I articulated why design of Multimodal Environmental Interface is an inherently Cross-Disciplinary Problem that should engage designers and artists along with Computer Scientists, Electrical Engineers, and Mechanical Engineers.

- I used the proposed framework to conduct user experiments.
- I did not prove my hypothesis that whereas discrete changes are easier to control with speech, continuous changes are easier to control with gestures. I proved my hypothesis that the perception of whether a gesture or a speech command feels intuitive is consistent among the majority of users.
- I made a comprehensive overview of the history and current trends in multimodal human-computer interaction and interactive design in art and architecture.
- I speculated on the future of Multimodal Environmental Interfaces.

Chapter 4

Conclusions and Future Work

4.1 Summary and Future Work

Dynamic and responsive environments are becoming increasingly more commonplace. Examples ranging from Smart Homes to Interactive Art projects have successfully demonstrated the advantages that such environments have to offer. These advantages range from improved health and ergonomics to novel culturally meaningful experiences. However, despite their high potential, dynamic responsive environments are still rare. This is partly due to the challenges in their implementation, the state of currently available technologies, and the numerous safety precautions required.

This thesis claims that if Spatial Interactive Environments are to become commonplace in the future, then we will need a framework for how to design and evaluate such environments. In this thesis, I proposed a way of thinking about spatial changes as *discrete* and *continuous*. I claimed that in order to understand how natural modes of interaction can be meaningfully related to spatial changes, we should conduct experiments with users. I outlined ways of thinking about spatial changes, built prototypes, and conducted user experiments. These three elements together form a basis for a framework for design and evaluation of MEI. The design of each of the experiments, however, was limited by the available technology, cost of equipment, time and the level of my technical expertise. I believe further modified versions of the experiments

need to be conducted to verify the current results. For example, my hypothesis that whereas discrete changes are easier to control with speech, continuous changes are easier to control with gestures proved to be wrong. However, it might be the case that imperfectly implemented gestural interaction significantly impacted the ways users perceived discrete and continuous tasks. Another factor was the size and age range of the user group: I chose ten graduate students between the ages of 22 and 36 with a good grasp of technology. If I were to take the experiment further, I would increase the test group size and its diversity to include users with both technical expertise and those without such expertise.

If we are to interact with our spatial environments in ways that are similar to those in which we interact with each other, then we need to gain a better understanding into how to relate spatial changes to natural modes of expression. This understanding can only come from designing different kinds of interactions and conducting user experiments. The thesis defines a systematic way of analyzing the relationship between changes and modalities; however, it is only a first step in what could become a framework for designing for change.

Appendix A

Tables

Table A.1: An overview of Interactive Art and Architecture

COMPANY	WORK
Art + Com	http://www.artcom.de/en/projects/project/detail/kinetic-sculpture/
Philips	http://www.lighting.philips.com/main/led/oled
Random International	http://random-international.com
dECOi	http://www.decoi-architects.org
Openended group	http://openendedgroup.com
SOSO	http://sosolimited.com
Hypersonic	http://www.hypersonic.cc
Patten Studio	http://www.pattenstudio.com
Plebian Design	http://plebiandesign.com/projects
Bot and Dolly	http://www.botndolly.com/box
IDEO	http://www.ideo.com/expertise/digital-shop
Zaha Hadid	http://vimeo.com/69356542
Kollision	http://kollision.dk
Cavi	http://cavi.au.dk
Troika	http://www.troika.uk.com/projects/
Nexus	http://www.nexusproductions.com/interactive-arts
Universal Everything	http://www.universaleverything.com
Dan Roosegaarde	http://www.studioroosegaarde.net
Minimaforms	http://minimaforms.com
RAA	http://www.raany.com
Onionlab	http://www.onionlab.com

Appendix B

Figures

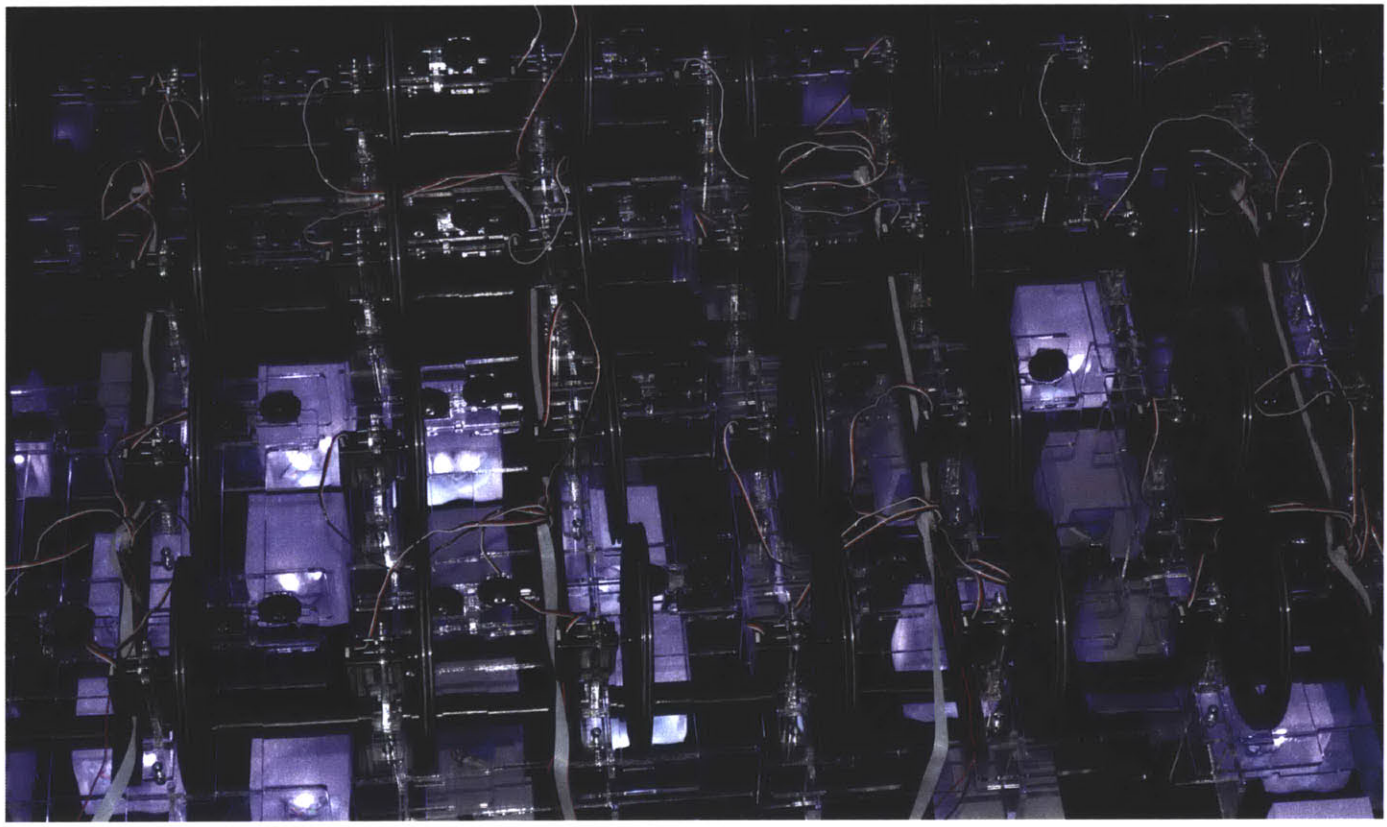


Figure B-1: Transformative Space Installation: view from top.

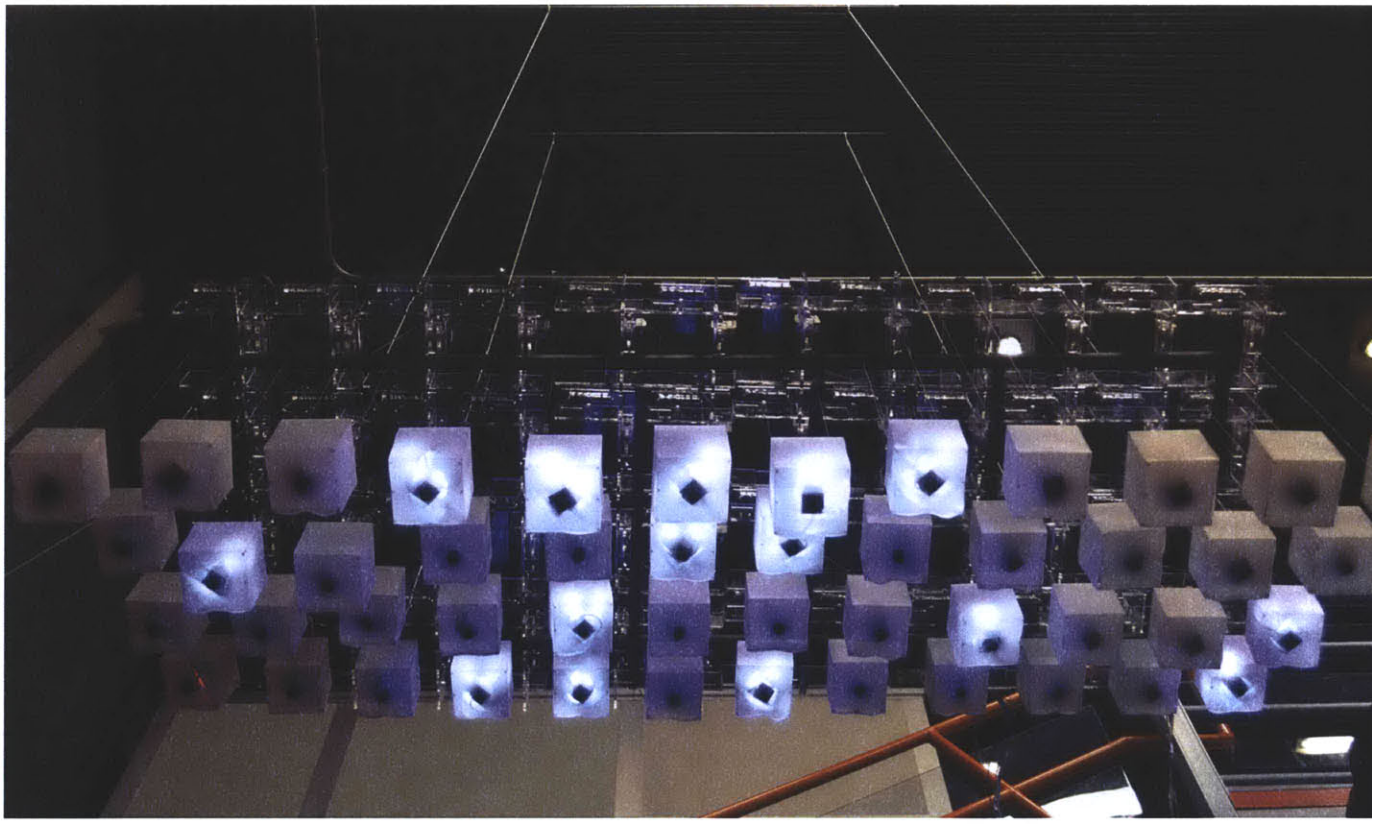


Figure B-2: Transformative Space Installation: view from below.

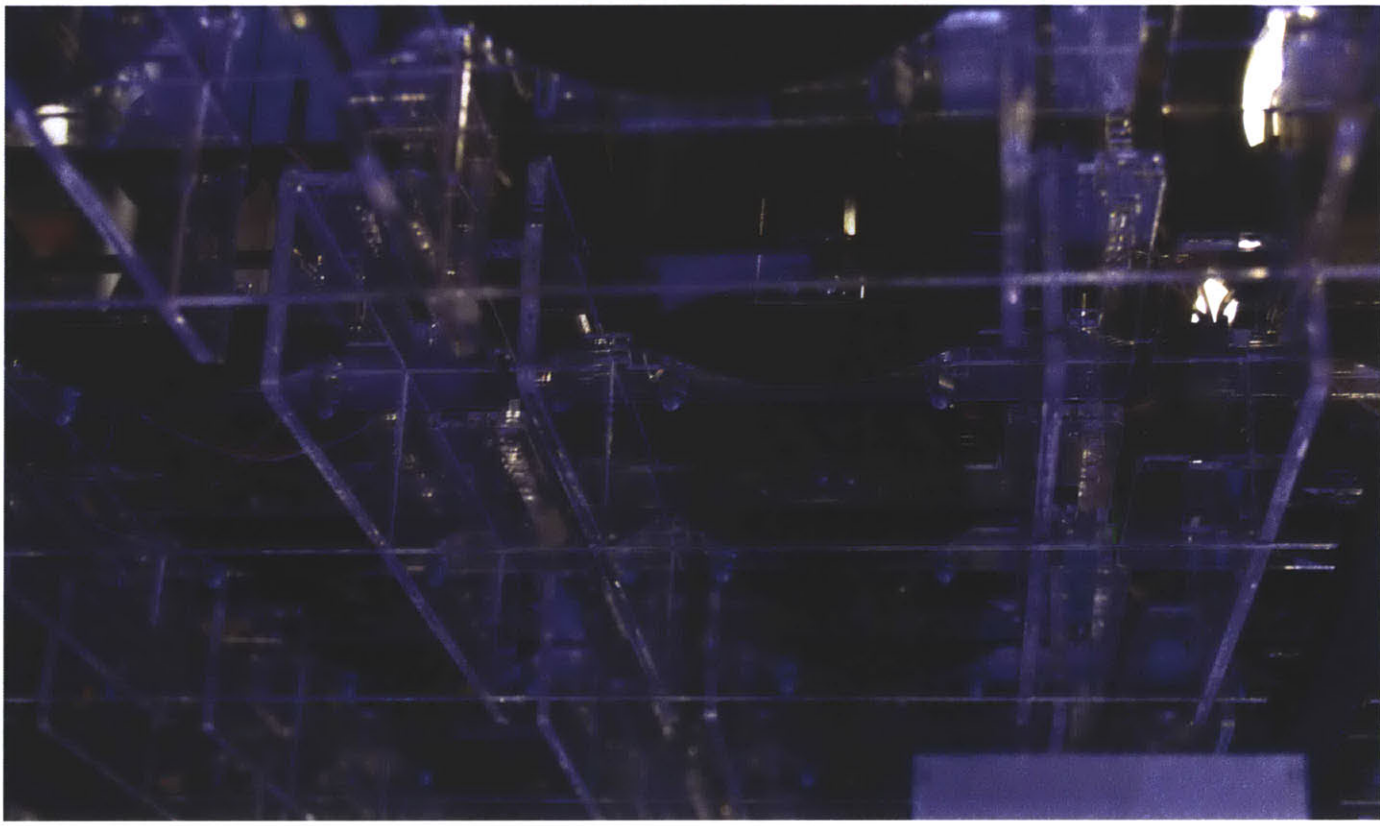


Figure B-3: Transformative Space Installation: close-up view.

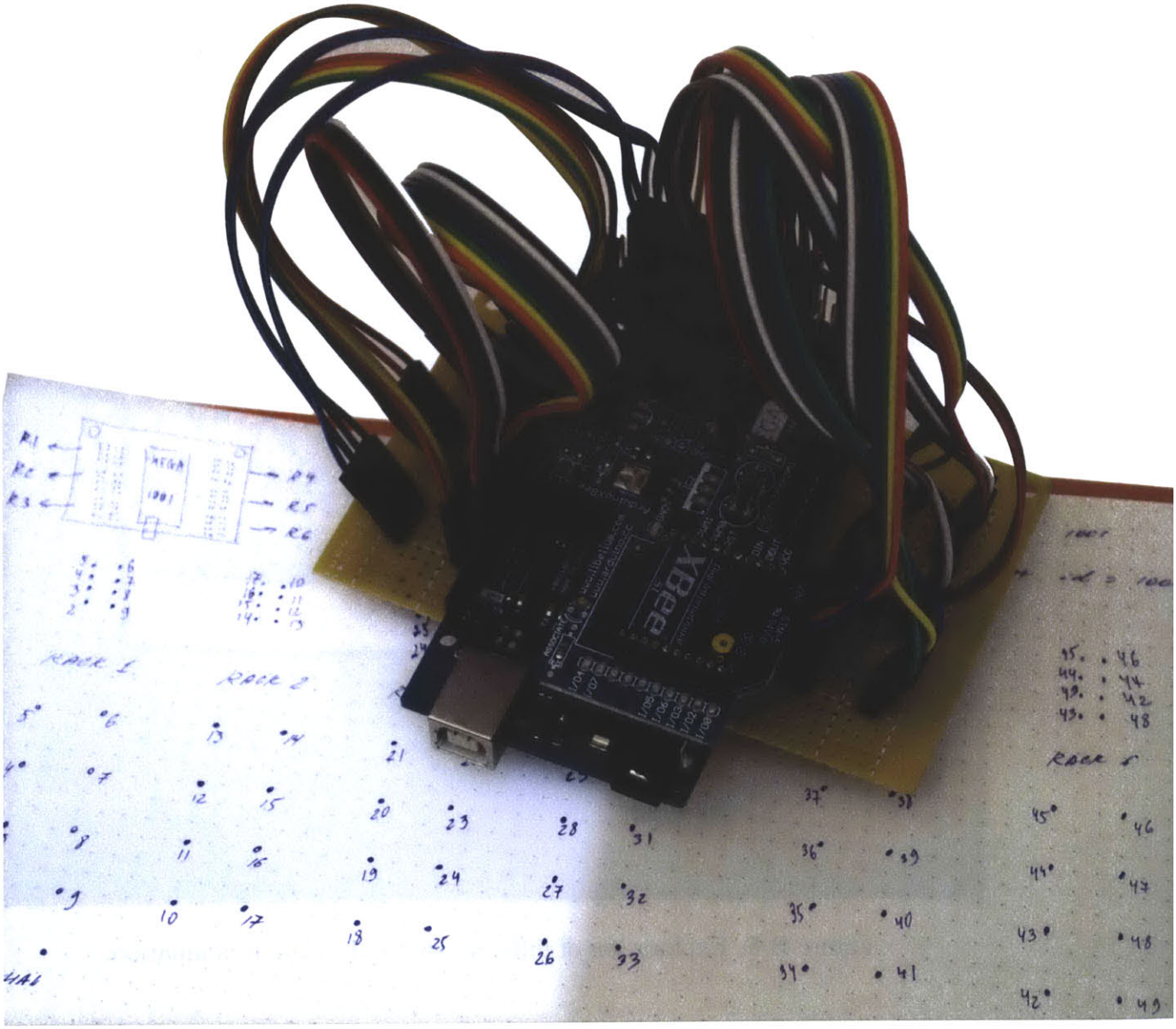


Figure B-4: Prototype Hardware: arduino mega board used power 48 servo motors.

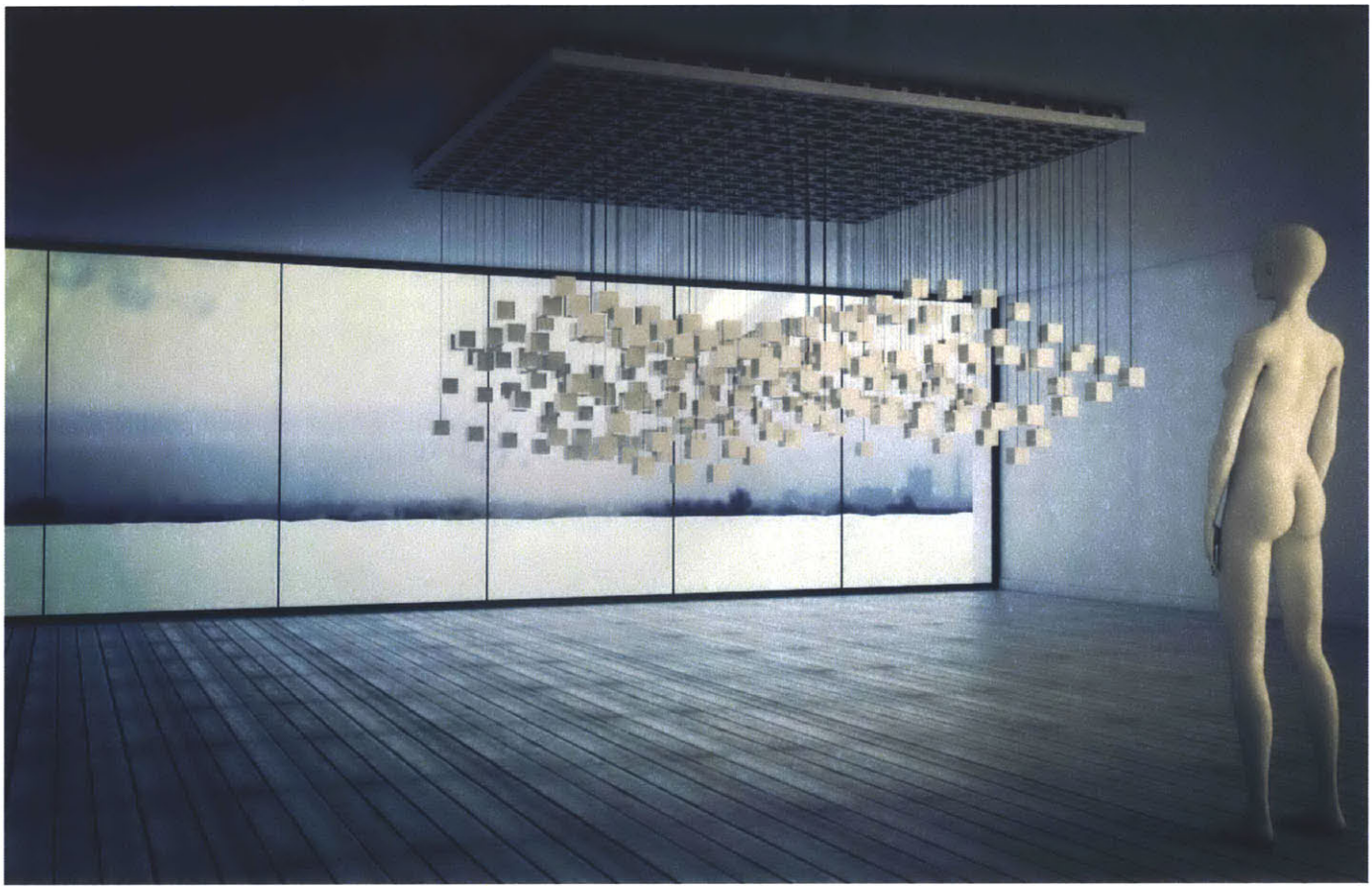


Figure B-5: Exploration of different states through an animation.



Figure B-6: An animated representation og computational reading of facial expressions.

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